Project #1: Sales Data Analysis

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I. Define the Problem or Question \

[What is the purpose of the analysis]

Ans: The purpose of this analysis is to see the trends, top products, customer type, reason behind the sales, gross margin and regions where most sales were made.

[Who is this analysis for ?]

Ans: This analysis is for the owner of the chain stores and it is also the governance team.

[Why is it important?]

Ans: This analysis is important for the owner because trough this analysis, he can determine the required changes that should be done in order to increase the revenue and customer interaction.

II. Detailed Analysis

[How did you approach your analysis, what techniques did you use]

Collect Data

[Summary of key variables and what they mean]

Clear Environment

Load Libraries and Custom Functions

Load Data

Clean Data

Convert Data to Useful Definitions

Modify Data - Create Calculated Fields

We have unit level data (price and cost) and aggregate revenue (line_total) and profit (gross margin) but are missing aggregate cost and profit margin %.

1. Group data by category and show sum of revenue, cost, gross margin, and average gmpcnt.

category	sum_revenue	sum_cost	sum_gm	sum_discount	avg_discpcnt <- mean(discount_pct)	avg_gmpcn
Beverages	62918.667	35762.5	27156.167	6902.1225	0.05152812	0.4631132
Hardware	39238.472	19843.5	19394.972	3956.7675	0.04651442	0.5178189
Household	26915.810	14484.8	12431.010	2533.1800	0.04721670	0.4952461
Housewares	3860.955	2064.0	1796.955	437.3250	0.05116279	0.4908708
Personal	4195.057	1807.6	2387.457	404.6925	0.04734513	0.6026533
Snacks	23715.295	10956.6	12758.695	2259.4750	0.04853229	0.5526811
Stationery	2043.608	833.8	1209.808	227.1825	0.05416667	0.6559702

Group by Store and save in a variable. Create a table and save the table

```
# A tibble: 10 × 7
   store_city  sum_revenue sum_cost sum_gm sum_discount avg_discpcnt <- mean(d…¹
   <fct>             <dbl>    <dbl>  <dbl>        <dbl>                    <dbl>
 1 Atlanta           9359.    4884.  4475.         959.                   0.0445
 2 Boston            8428.    4385.  4043.         805.                   0.0496
 3 Chicago          30113.   15892. 14221.        3157.                   0.0492
 4 Dallas            9580.    4976.  4603.         838.                   0.0415
 5 Denver            7857.    4115.  3741.         781.                   0.0531
 6 Los Angeles      31084.   16449. 14635.        3290.                   0.0508
 7 Miami             7973.    4099.  3874.         805.                   0.0480
 8 New York         43772.   23289. 20483.        4768.                   0.0525
 9 Phoenix           7251.    3836.  3415.         678.                   0.0445
10 Seattle           7471.    3826.  3645.         640.                   0.0422
# ℹ abbreviated name: ¹`avg_discpcnt <- mean(discount_pct)`
# ℹ 1 more variable: avg_gmpcnt <dbl>
```

Explore Data

Evaluate your overall data set using SKIMR

Here is a checklist of things to check on when evaluating data:

1. Check the structure and metadata - Evaluate how many rows, columns, data-types, and so forth. Ans: There are 2500 rows and 18 column in sales date and after cleaning up the data by changing the variable type and making some necessary column by using mutate function now it has 2500 rows and 23 column.

2. Missing Data - Evaluate if your data is complete and identify missing data overall and, if so, particular patterns (e.g. “monday transaction data always is missing cost”)

3. Data Quality & Consistency - What are the ranges of values and does the data make sense? Are data elements consistent - e.g. is “M”, “Male”, and “Men” values in the same field

4. Relationships and Distributions - Evaluate the summary statistics - mean, std. deviation, min, max, etc. - and the distribution of the data. Is it normally distributed?

5. Test of Reasonableness (TOR) - Does the data make sense in the context of the analysis?

SKIMR provides a quick way to look at all the key aspects of your data with a single command. The first window looks at your overall data set and then each subsequent window looks at the variables based on type (chr, factor, date, etc.).

Data summary

Name	sales_data_clean
Number of rows	2500
Number of columns	23
_____	
Column type frequency:	
character	4
Date	1
factor	8
numeric	10
_____	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
transaction_id	0	1	11	11	0	2500	0
transaction_datetime	0	1	19	19	0	2498	0
store_id	0	1	4	4	0	10	0
product_id	0	1	4	4	0	25	0

Variable type: Date

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date	0	1	2025-05-01	2025-08-31	2025-07-02	123

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
store_city	0	1	FALSE	10	New: 622, Los: 502, Chi: 459, Dal: 177
store_region	0	1	FALSE	4	Wes: 843, Nor: 744, Mid: 459, Sou: 454
product_name	0	1	FALSE	25	Esp: 375, Toi: 352, Lux: 346, USB: 332
category	0	1	FALSE	7	Bev: 818, Sna: 511, Hou: 503, Har: 416
promo_code	0	1	FALSE	5	NON: 1403, SUM: 475, VIP: 282, WEE: 221
payment_method	0	1	FALSE	4	Cre: 1492, Cas: 403, Deb: 370, Mob: 235
customer_type	0	1	FALSE	2	Ret: 1396, New: 1104
loyalty_member	0	1	FALSE	2	Tru: 1279, Fal: 1221

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
unit_price	0	1	14.45	6.46	2.49	11.99	13.99	18.99	25.99	
unit_cost	0	1	6.86	3.23	0.80	5.00	7.00	9.00	12.00	
quantity	0	1	5.02	5.27	-30.00	1.00	3.00	5.00	30.00	
discount_pct	0	1	0.05	0.06	0.00	0.00	0.00	0.10	0.20	
gross_margin	0	1	30.85	34.39	-194.70	9.98	19.98	41.97	317.26	
line_revenue	0	1	65.16	75.36	-449.70	15.99	37.98	85.45	617.26	
line_cost	0	1	34.30	42.04	-255.00	8.50	18.00	45.00	300.00	
gross_margin_pcnt	0	1	0.51	0.08	0.29	0.45	0.52	0.56	0.73	
unit_discount	0	1	0.71	1.07	0.00	0.00	0.00	1.30	5.20	
line_discount	0	1	6.69	12.22	0.00	0.00	0.00	9.50	129.95	

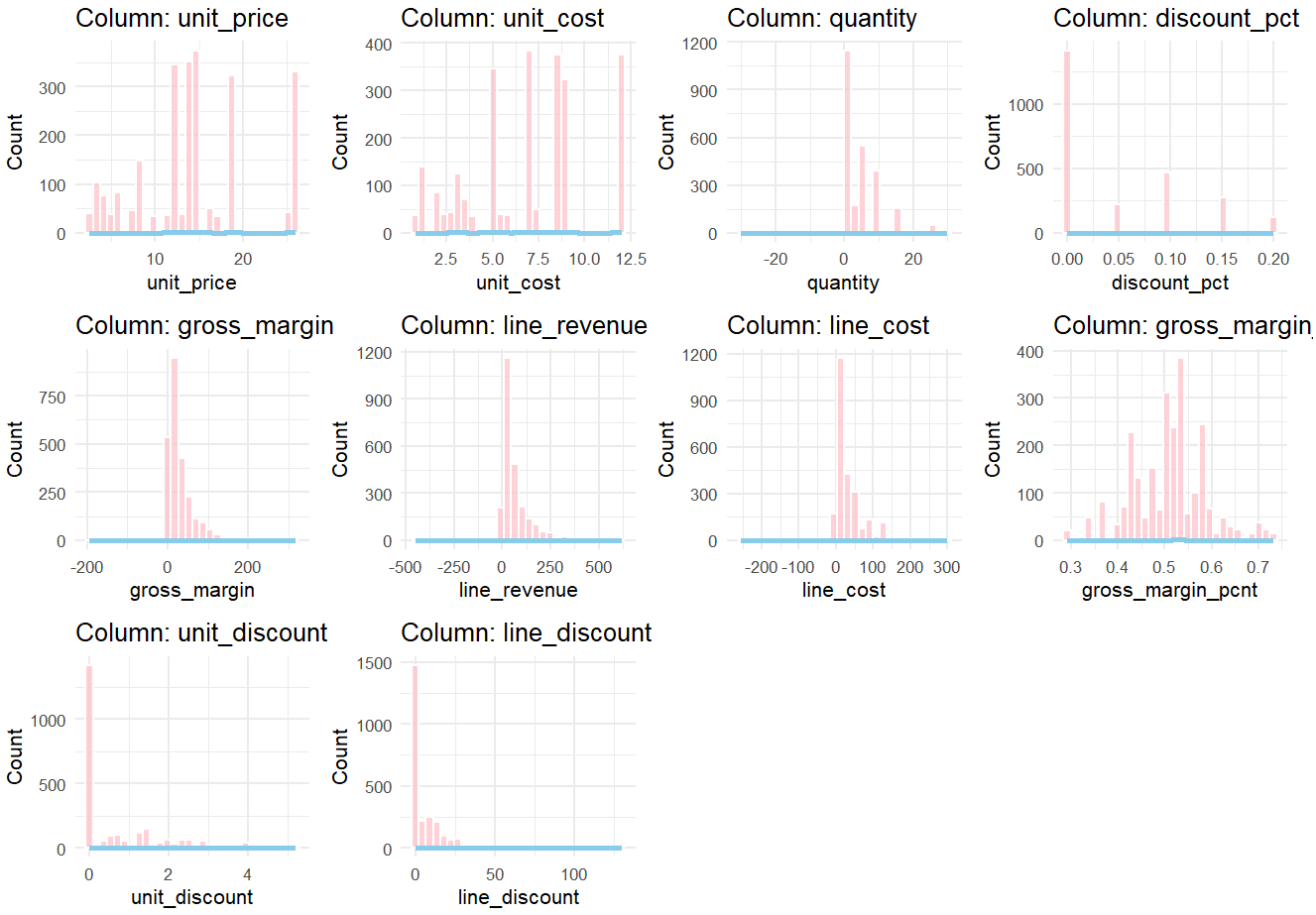
This is not a fun step but is important to do early in your analysis. Remember the GIGO principle - garbage in = garbage out. If you have bad data, no amount of brilliant analysis will over come it.

Exploratory Analysis Table - Build Table to Focus Analysis

The table helps the analyst ensure all the main information is analyzed. For each categorical variable, evaluate each numeric value. Judgement should be applied and unnecessary or unrelated data can be eliminated but it is important to ensure it is a conscious decision.

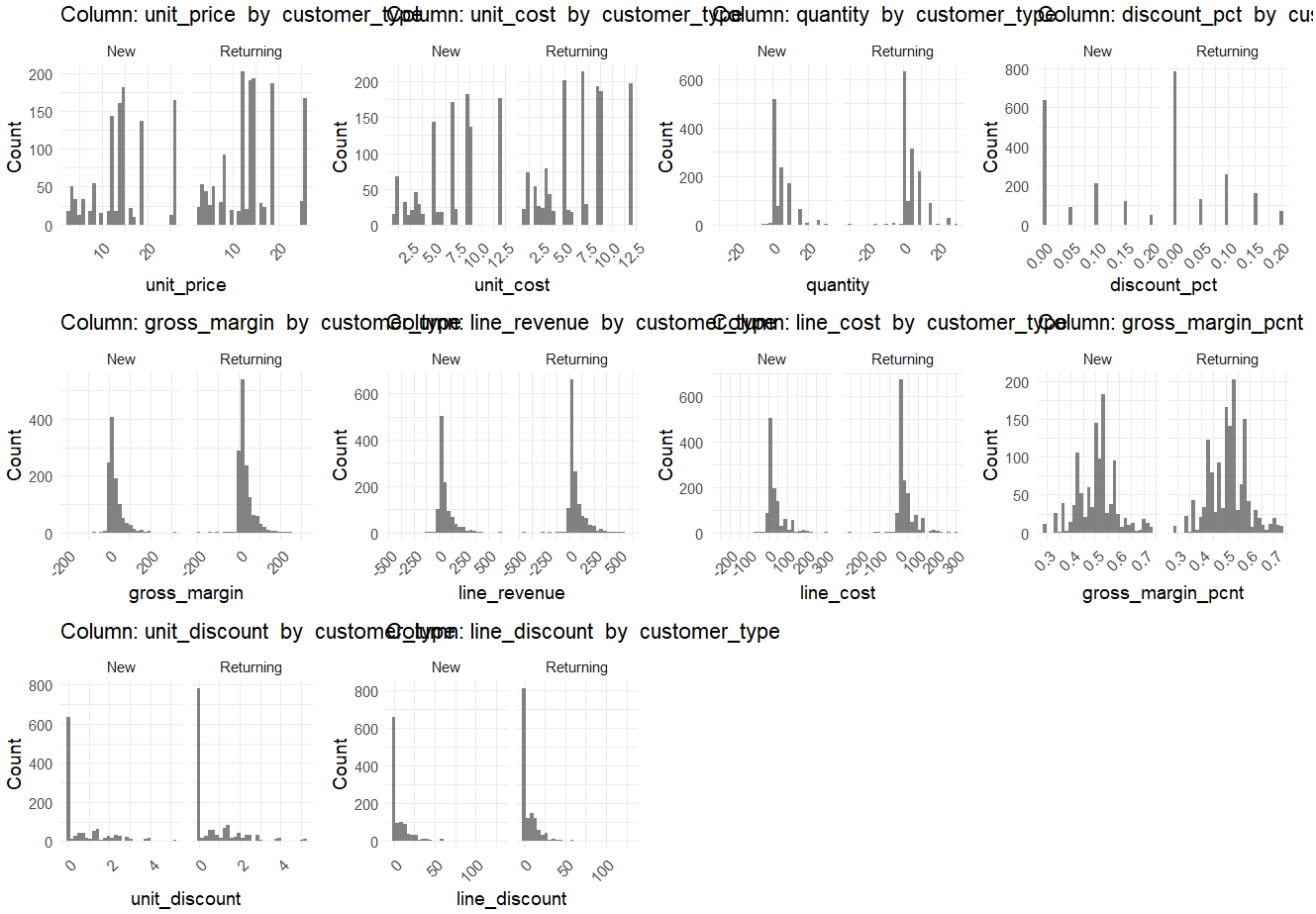
The table below lists the combination of categorical and numeric variables that should be analyzed.

Evaluate all numeric values to determine patterns in the data



Evaluate numeric values grouped by categorical values

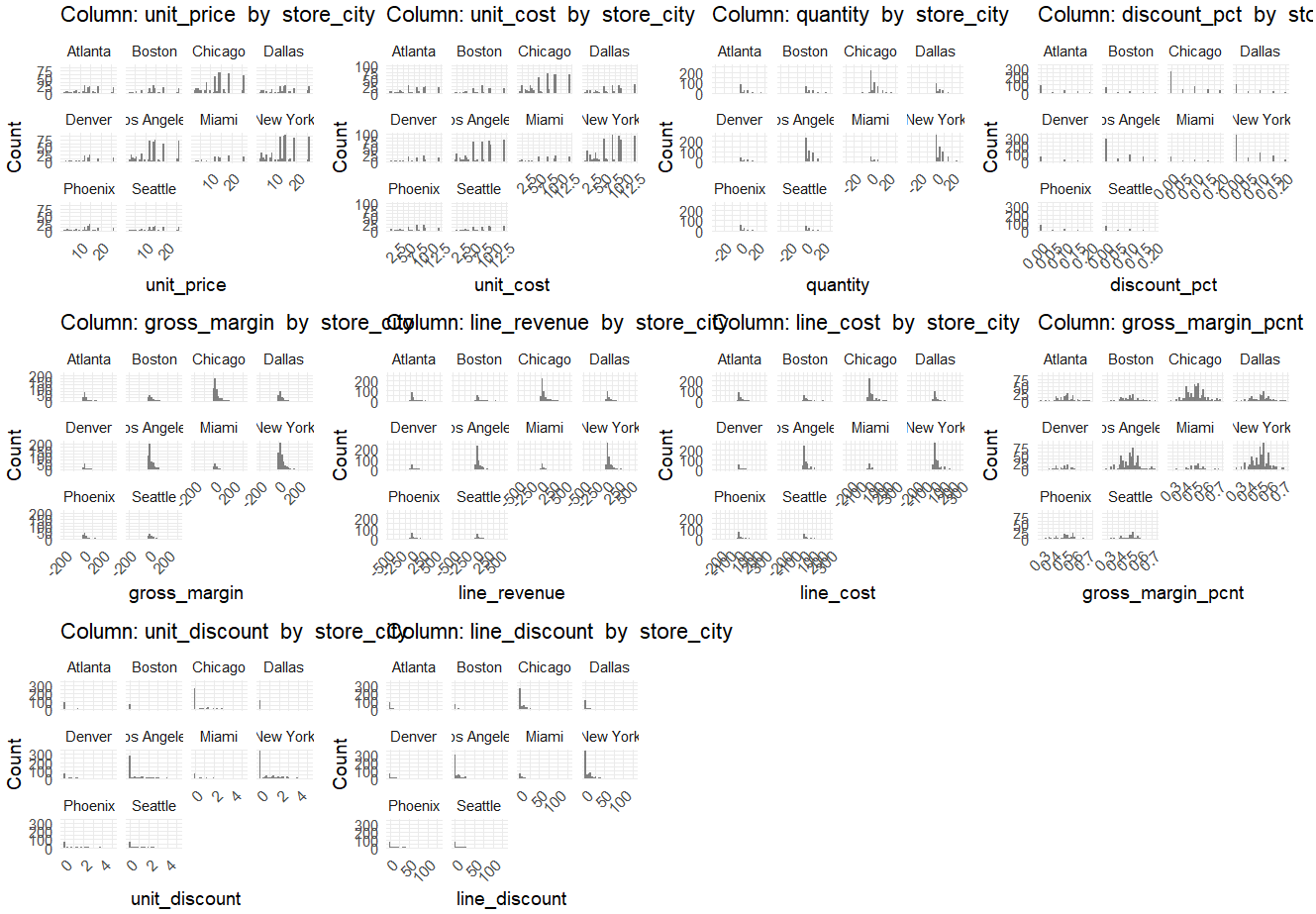
1. Evaluate all numeric data by customer type



Findings:

- Unit cost and unit price between new and returning customer is very similar. Between both groups, unit price is constantly higher than unit cost, which indicated stable markup structure.
- Returning customers shows more gross margin percent than new customers. It indicates returning customers can bring comparatively more profit per transaction.
- Most sales come from small purchases, though a handful of big ones push total income up—true for both fresh and repeat buyers. Revenue figures tilt toward lower amounts, yet spike because of rare expensive orders. Bigger deals pop up rarely. The pattern holds steady across customer groups, leaning right on the scale each time.

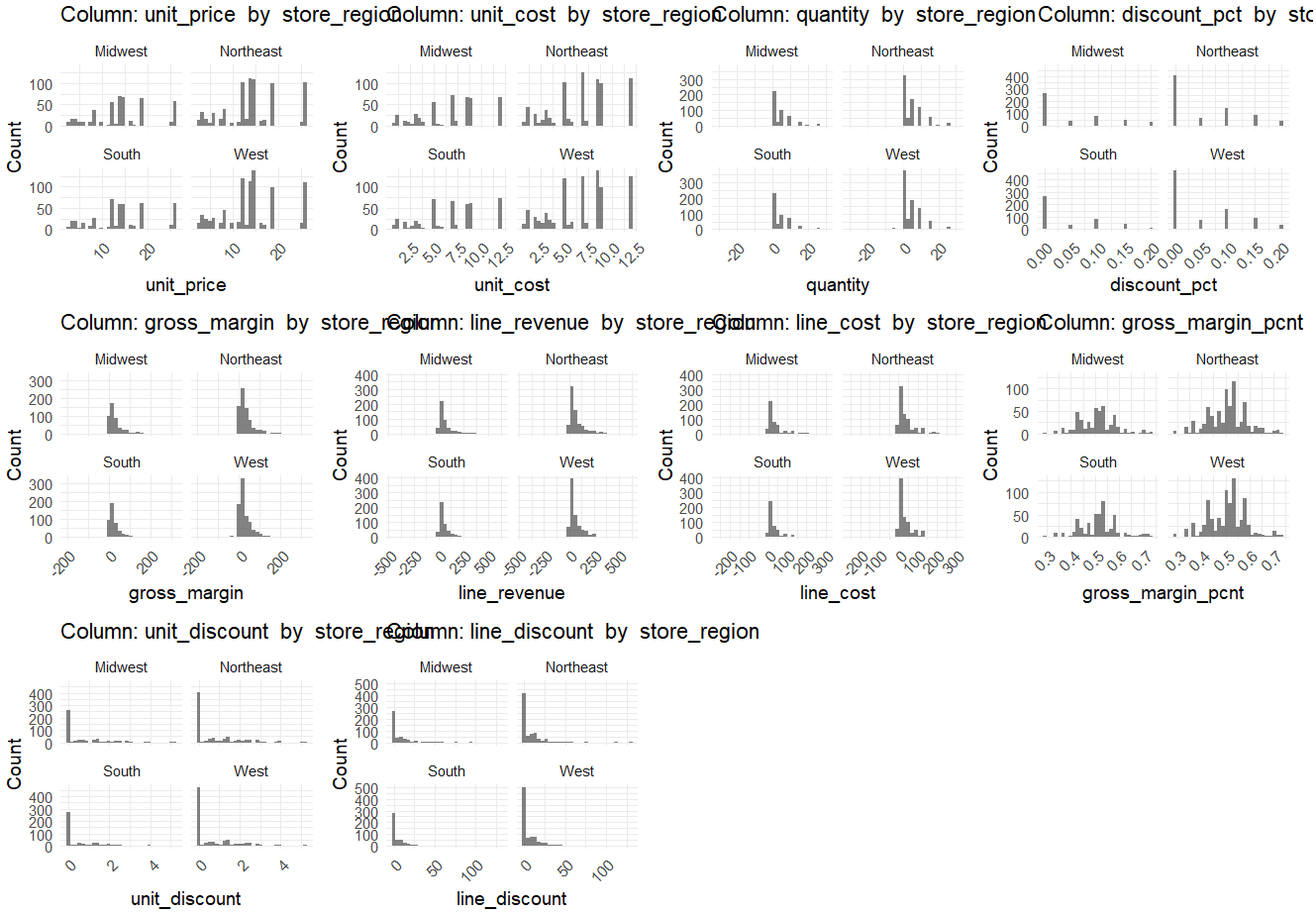
2. Evaluate all numeric data by store city location



Findings:

- Every city, line revenue and line cost are skewed right. This means most of the transaction are happening are of low prices.
- With gross margin and line revenue, Chicago, Los Angeles, and New York shows more deviation. It means there are more transaction and they have more sales volume compared to other cities.
- The gross margin percent distributions are similar across all the cities. Indicating a standardized pricing and profit ratio across markets.
- Most cities, the spread of prices and costs look alike. The price is always above the cost.
- In terms of quantity, the distribution is skewed right and this indicates most of the customers buy small value items.
- Discounts are always little to nothing. Occasional some discounts appear, which shows selective discount on specific items.

3. Evaluate all numeric data by store region

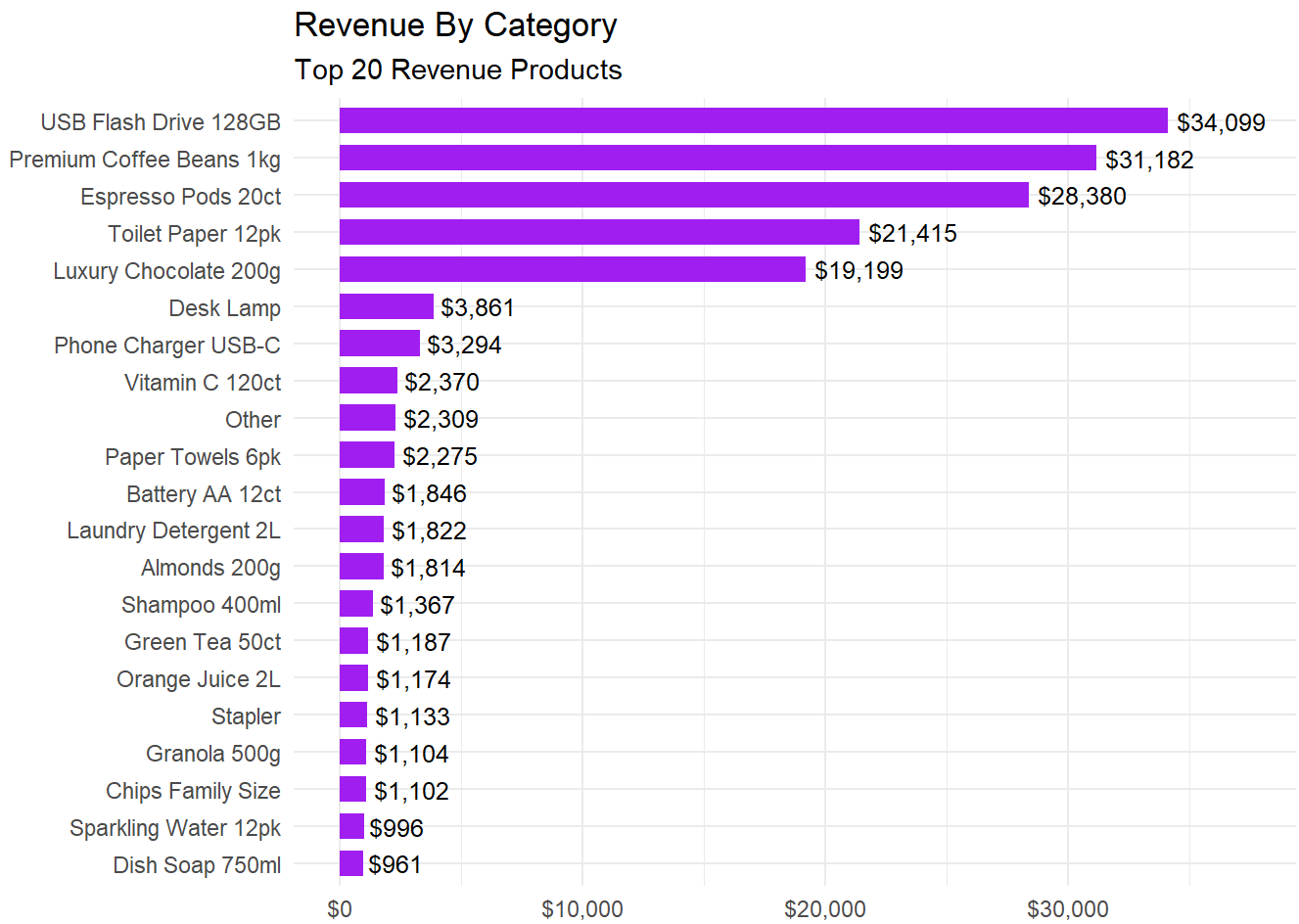


- For all the region, Revenue and cost are skewed right and it means transactions are small in value but higher in quantity.
- Gross margin percentages, are consistent across the regions, showing standardized pricing and profit ratio.
- All the regions, had little to no discount for most transaction.
- Unit price and unit cost are similar every where and it indicates a stable markup structure across regions.
- Quantity distribution is skewed right and it indicates that the most purchases are made are small and some in bulk.
- West and Northwest shows slightly better performance in GM. This is a indication on higher sales and stronger transaction.

4. Evaluate all numeric data by product category

Utilize bar charts to compare and rank numeric data by a category

1. Evaluate top 20 revenue generating products

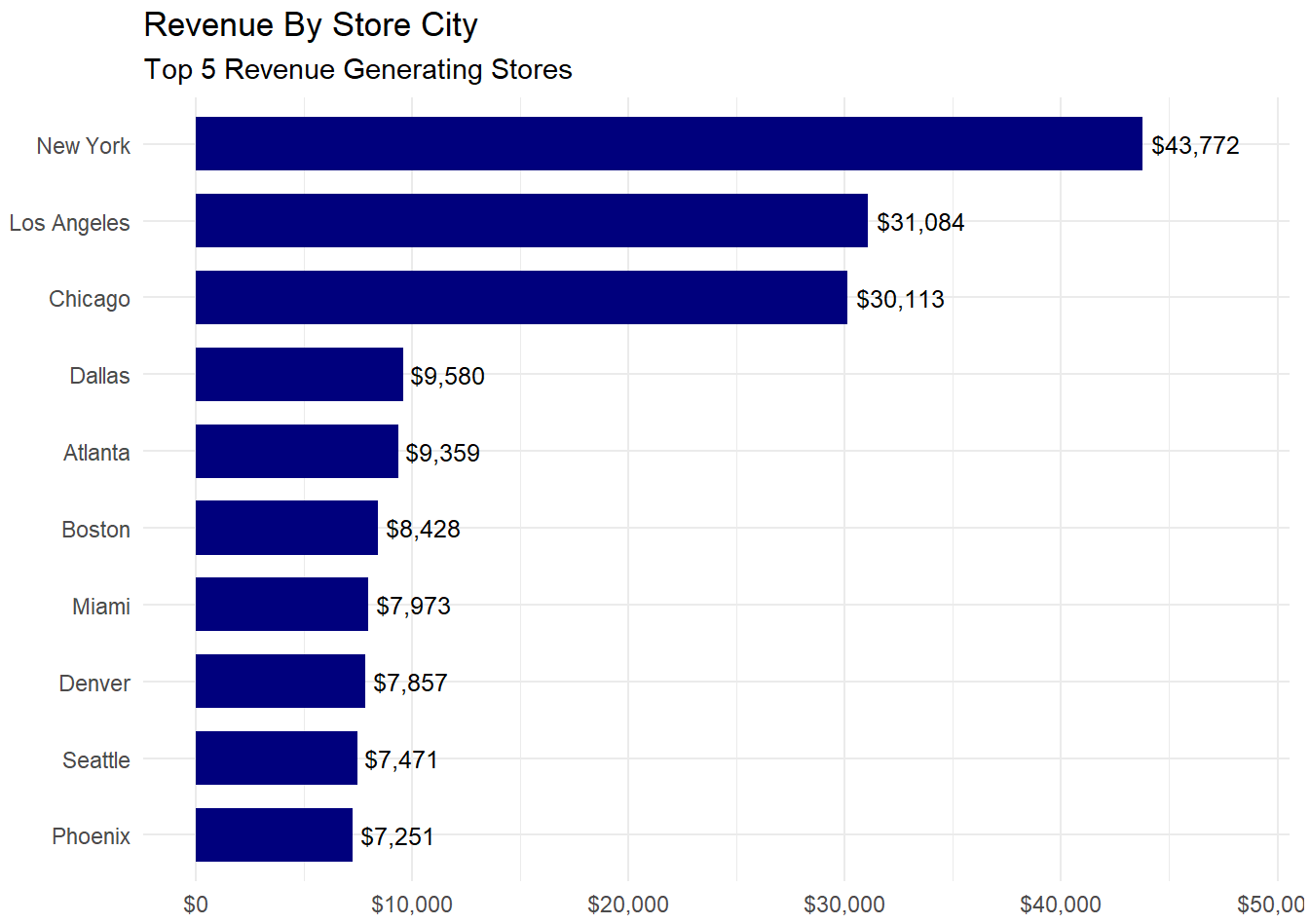


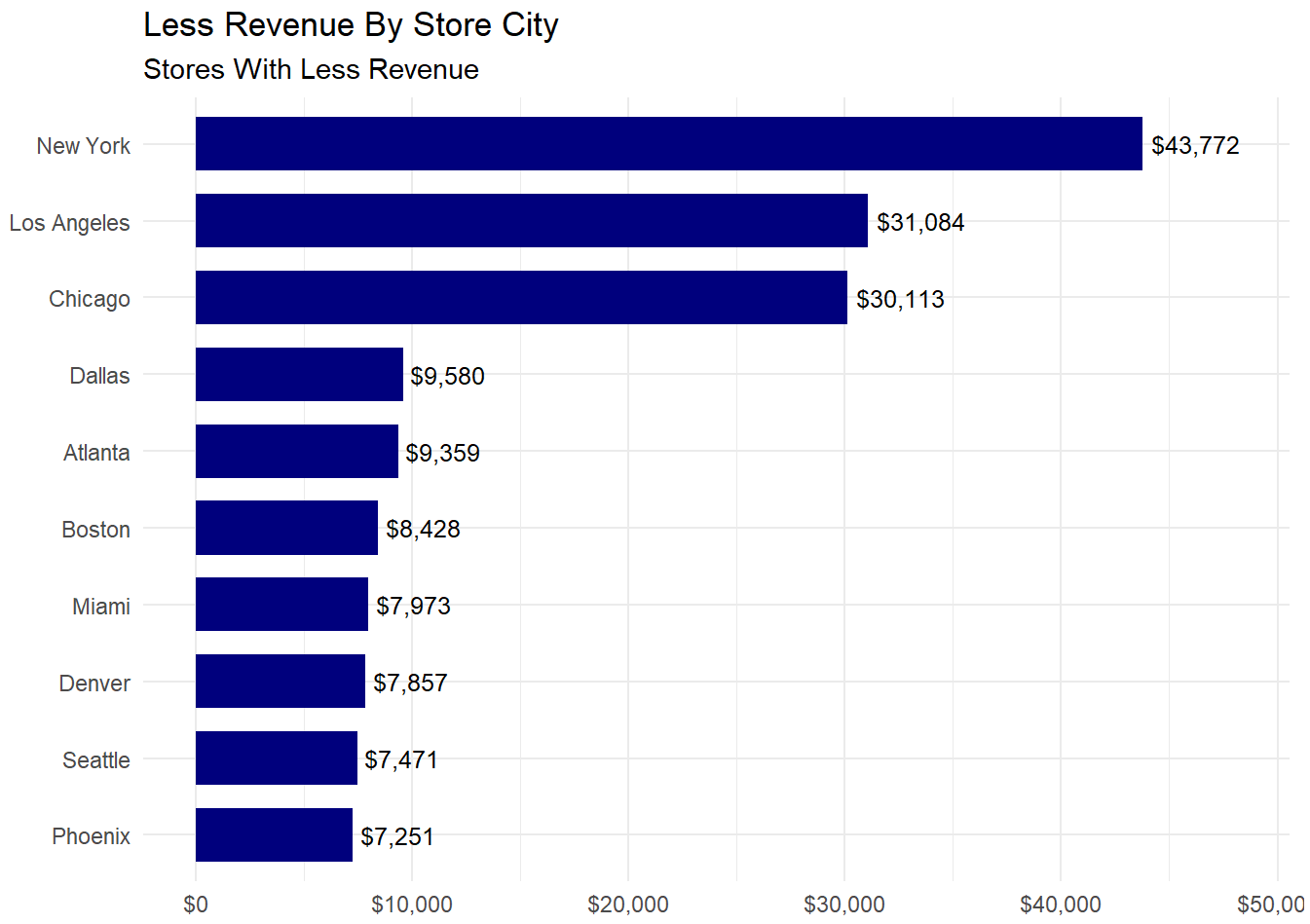

```
[1] "$134,274"
```

Findings:

- Top product by is USB Flash Drive 128GB which has $34,099.
- Top 5 is high % of total is $134274.

2. Evaluate top 5 revenue generating stores

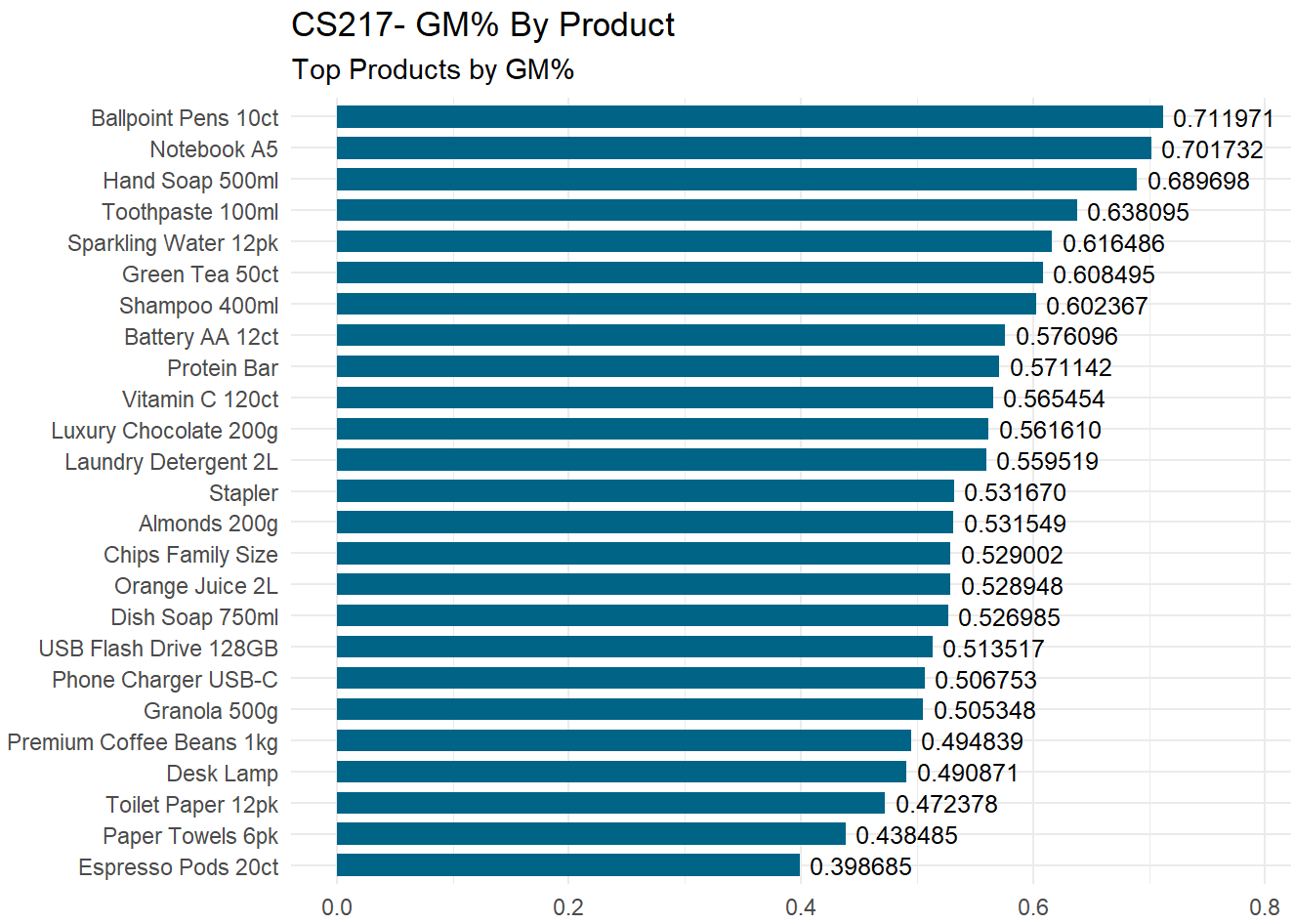

Revenue By Store City
Top 5 Revenue Generating Stores
New York
Los Angeles
Chicago
Dallas
Atlanta
Boston
Miami
Denver
Seattle
Phoenix
$43,772
$31,084
$30,113
$9,580
$9,359
$8,428
$7,973
$7,857
$7,471
$7,251
$0
$10,000
$20,000
$30,000
$40,000
$50,00



Findings

- The highest revenue generating stores by city was New York and they generated $43,772.
- Stores in Los Angeles and Chicago were after New York in terms of revenue, which is $31,084 and $30,113.
- Rest of the cities has transactions below $10,000.

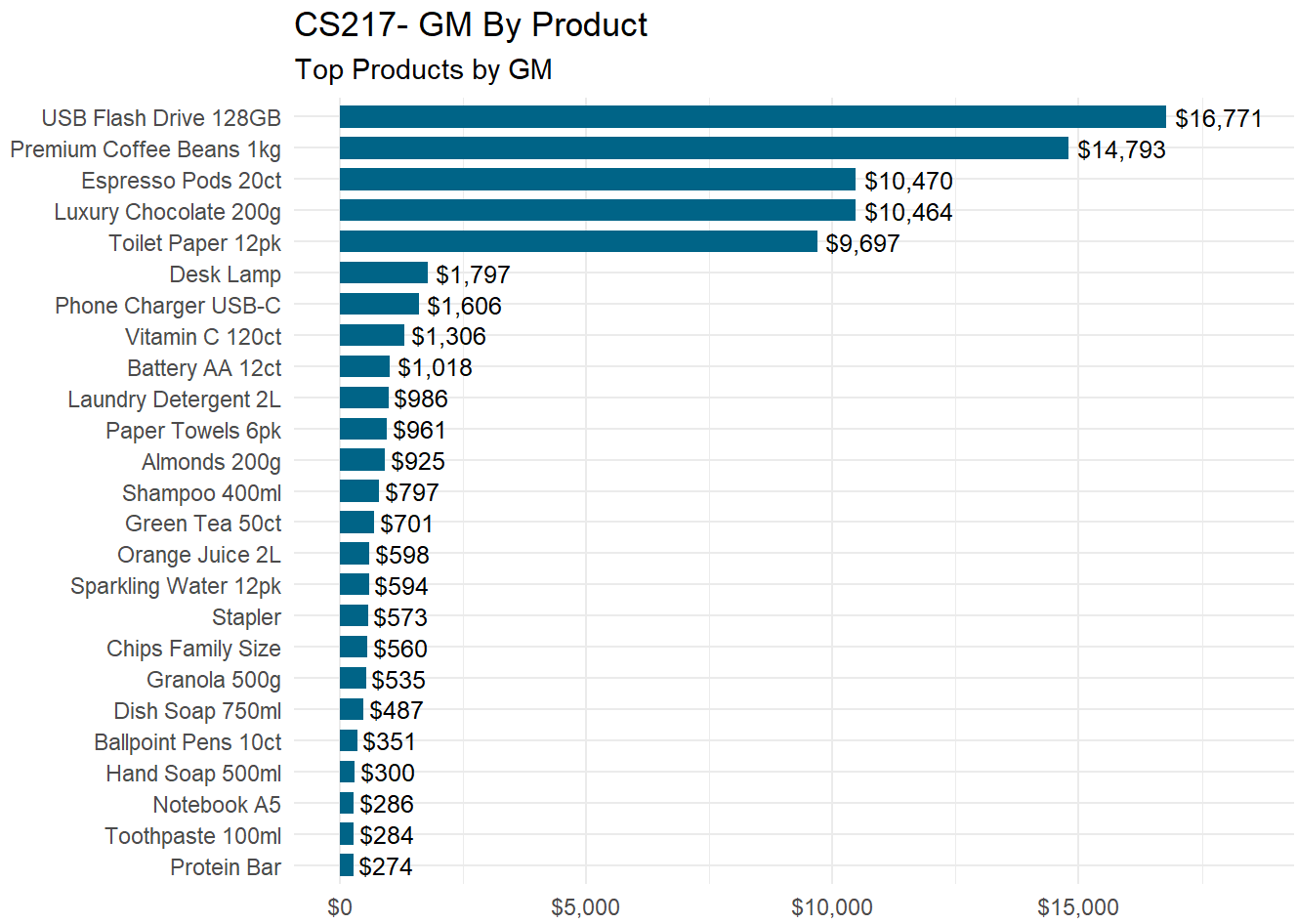
3. Evaluate top 25 most profitable products



Findings:

- Ballpoint Pens 10ct and Notebook A5 has the highest two gross margin percent 71% and 70%.
- Most products cluster between 50% to 60% GM%.
- Espresso Pods 20ct and Paper Towels 6pk has the least GM% , with 39% and 43% accordingly.

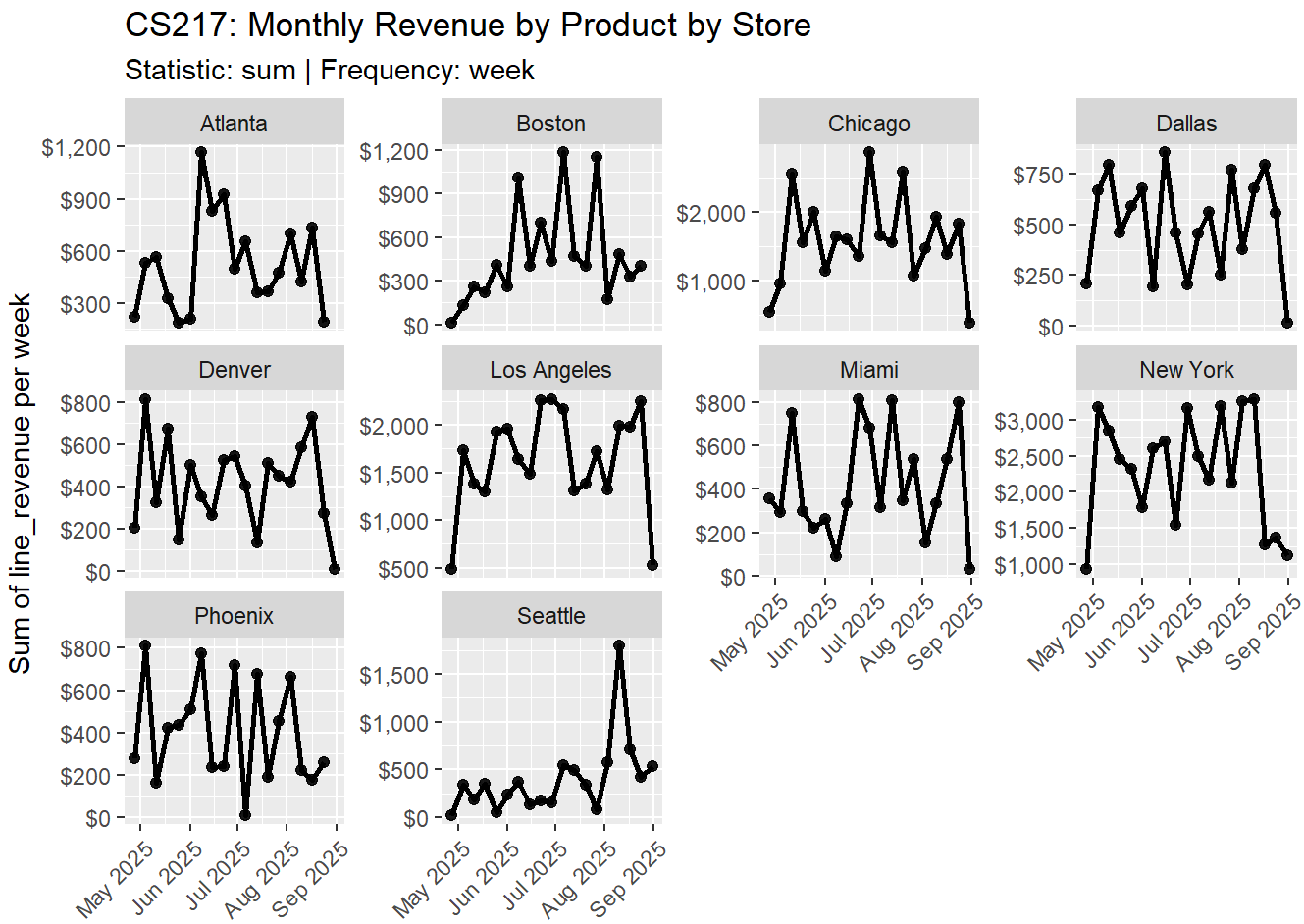
4. Evaluate top 5 most profitable product categories



- USB Flash Drive 128GB
- Premium Coffee Beans 1kg
- Espresso Pods 20ct
- Luxury Chocolate 200g
- Toilet Paper ### Perform a Trend Analysis

Perform a time series analysis. The key value of trend analysis is to see change in values over time (can be a date, age of a person, how long people are in line, etc.).

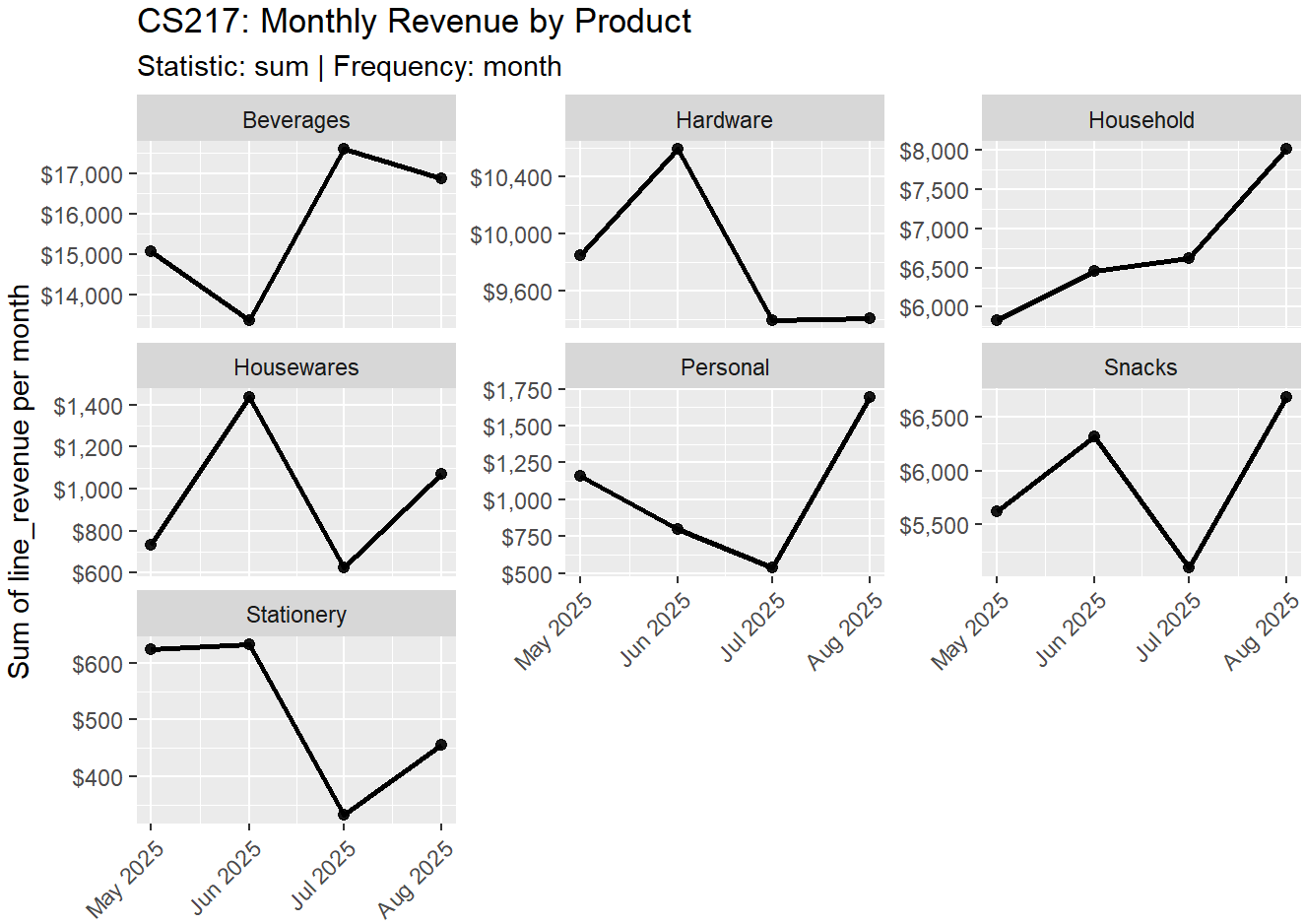
1. Evaluate monthly trends of revenue by month



Findings

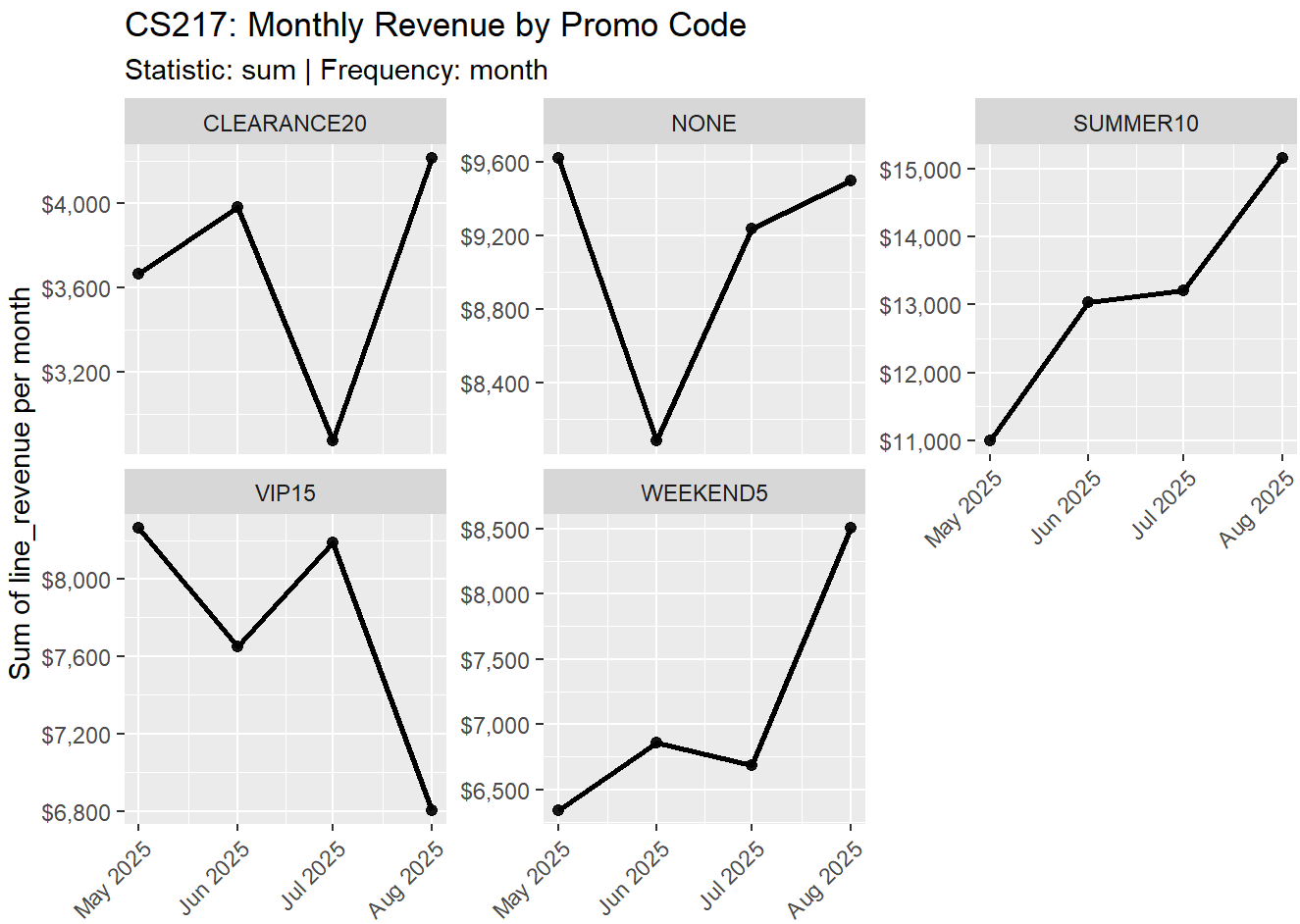
- New York has the highest amount of monthly revenue generated among all the cities.

2. Evaluate monthly trends of revenue by month by product category



Findings - Beverages generated most of profit then any other product. - During June and July, stationery items were lowest. Most probably for students getting summer break.

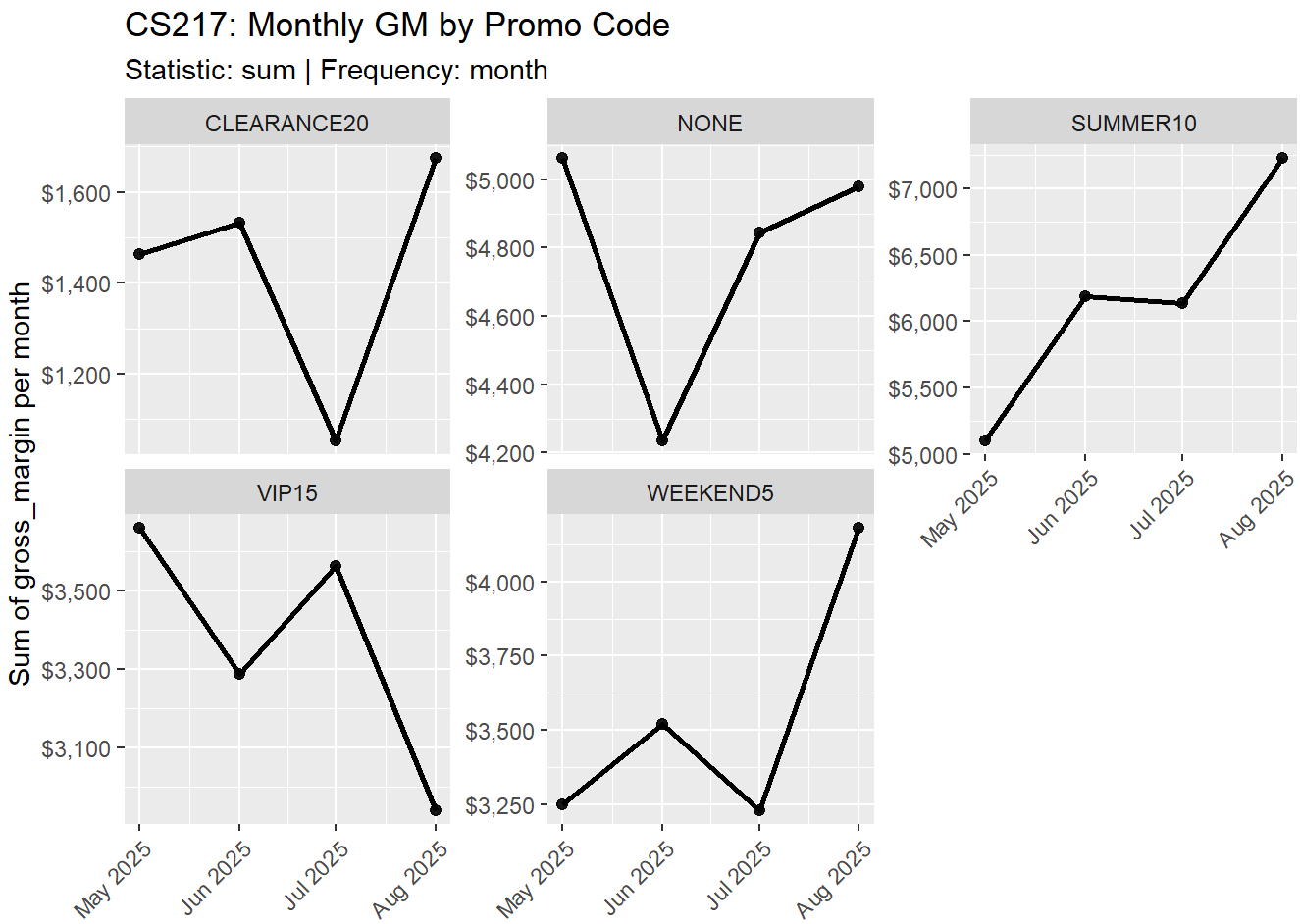
3. Evaluate monthly trend by month and any other category

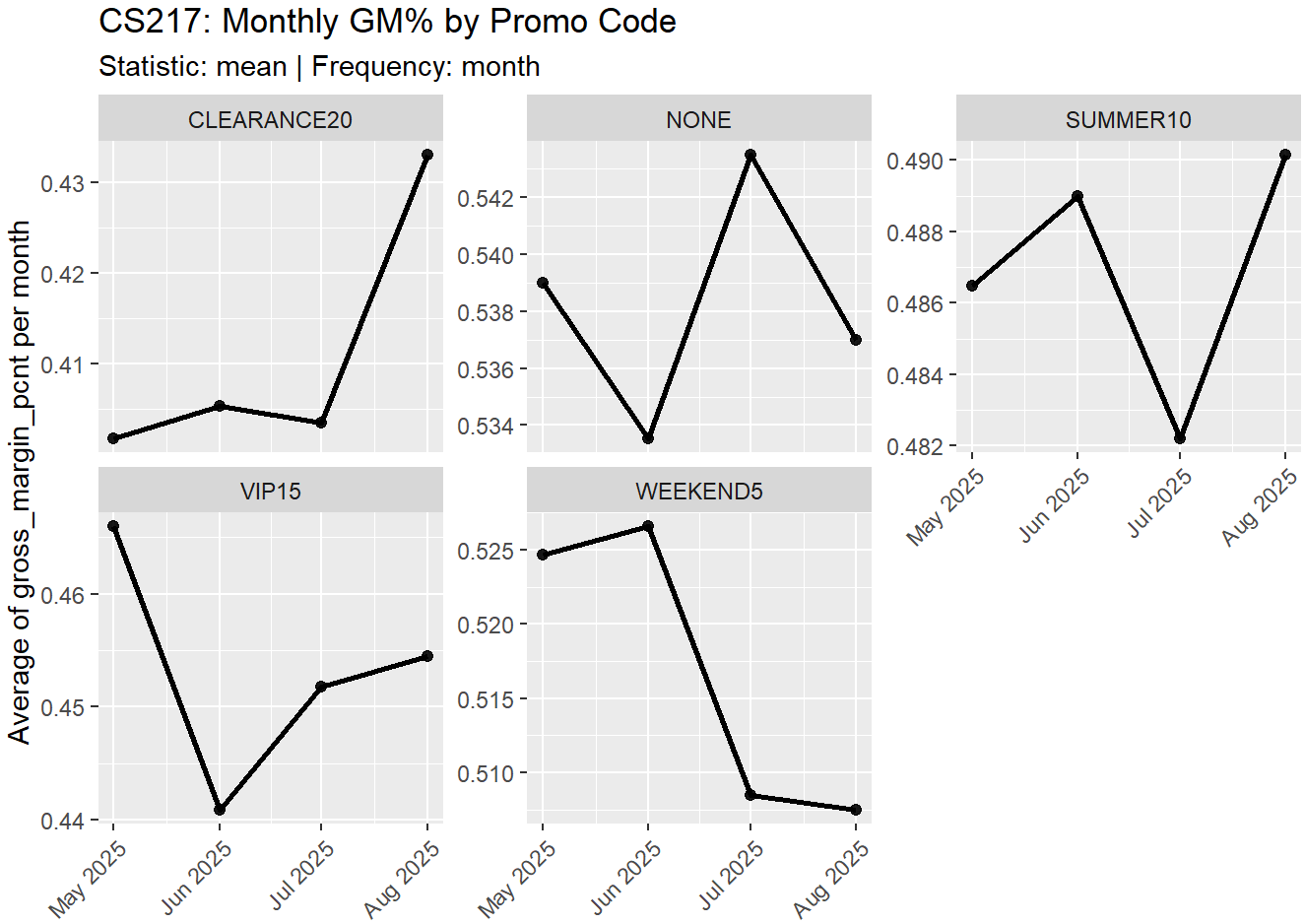


Finding

- Highest revenue made by promo code was SUMMER10 and CLEARANCE20 made the least revenue.

4. Look at Gross Margin $ by a key categorical variable


CS217: Monthly GM by Promo Code
Statistic: sum | Frequency: month
CLEARANCE20
NONE
SUMMER10
VIP15
WEEKEND5
Sum of gross_margin per month
$1,600
$1,400
$1,200
$5,000
$4,800
$4,600
$4,400
$4,200
$7,000
$6,500
$6,000
$5,500
$5,000
$3,500
$3,300
$3,100
$4,000
$3,750
$3,500
$3,250
May 2025
Jun 2025
Jul 2025
Aug 2025



Finding:

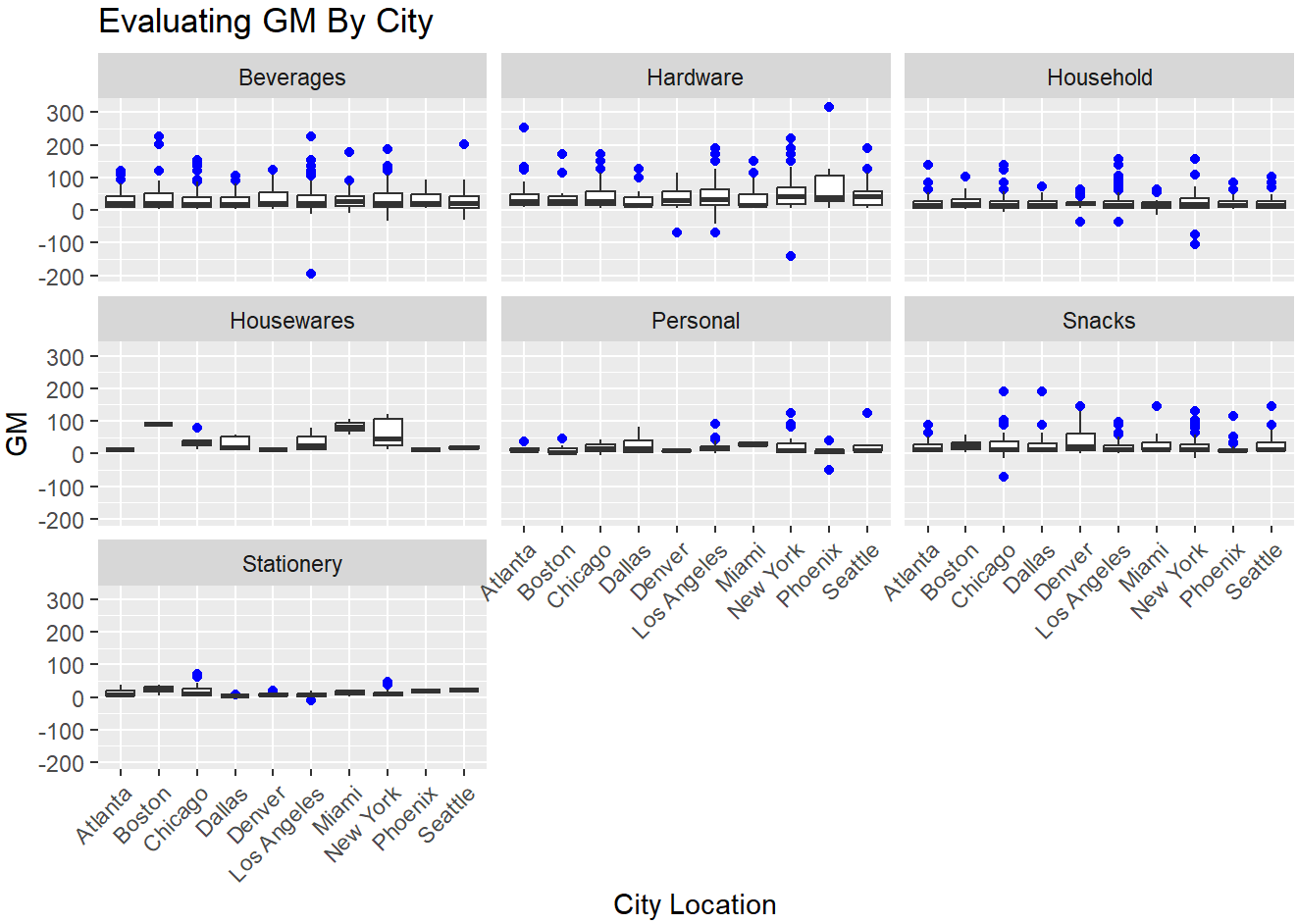
Graph 1

- The highest amount of GM was made by SUMMER10 promo code and it has a GM between $5000 and $7000, and CLEARANCE20 gave the least GM of around $1000-$1700.

Graph 2

- GM% wise, products with NONE promo code made more profit with the peak of approximately 54.5% and second place was promo with WEEKENDS. Here also CLEARANCE has the least profit provider by giving GM% of around 40%.

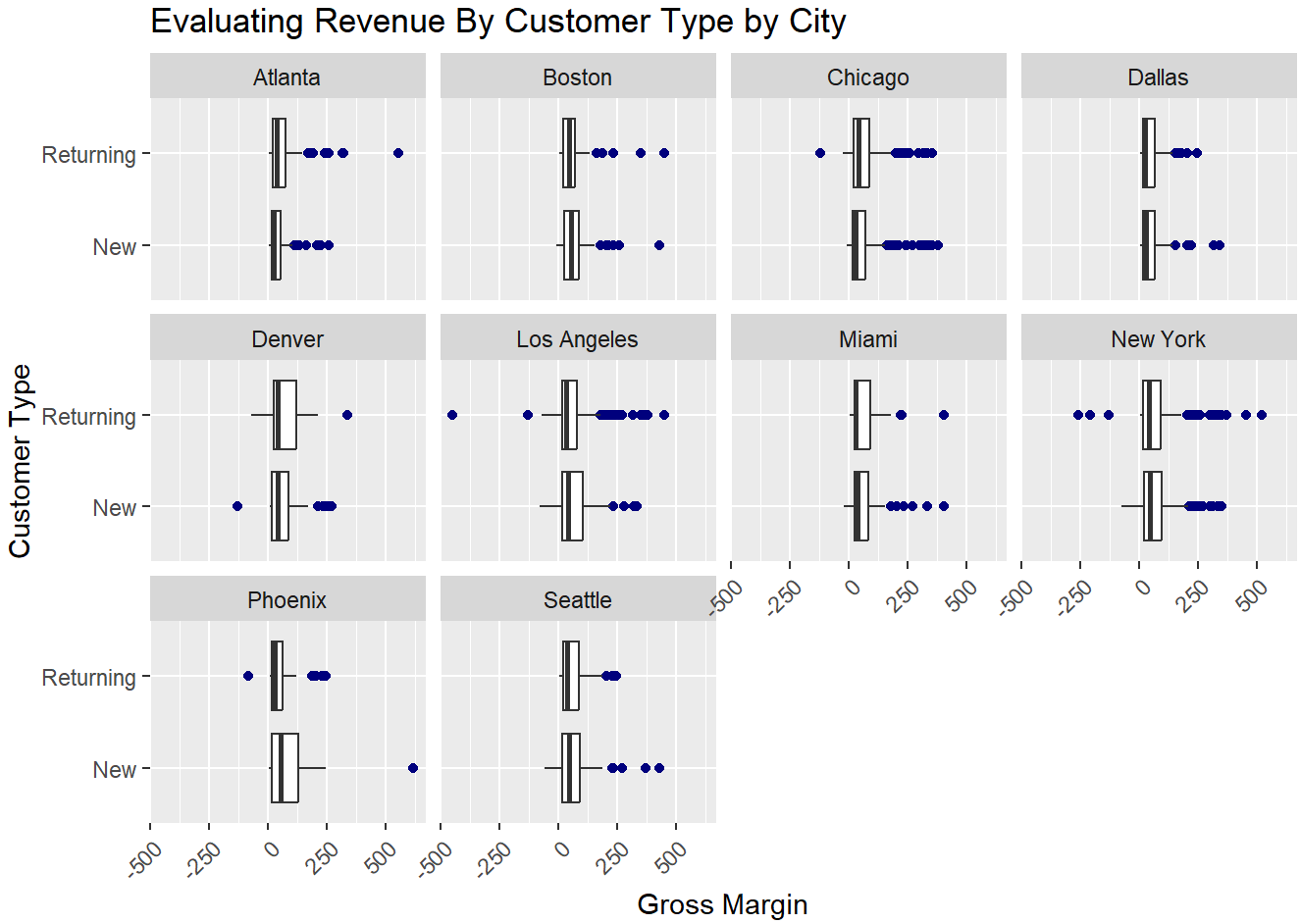
Utilize Box Plots to Evaluate Distribution of Revenue by City by Promotion Code



Finding:

- Across cities, Hardware has the largest spread and most outliers, which shows more transaction and profit.
- Household, personal, and stationary products are very stable in all cities. It indicates similar distribution across all cities and predictable gross margin.
- New York, Chicago, and Los Angeles has most spread and outliers troughout most of the product, and Phoenix and Denver shows a good transaction but not as good as the three cities.

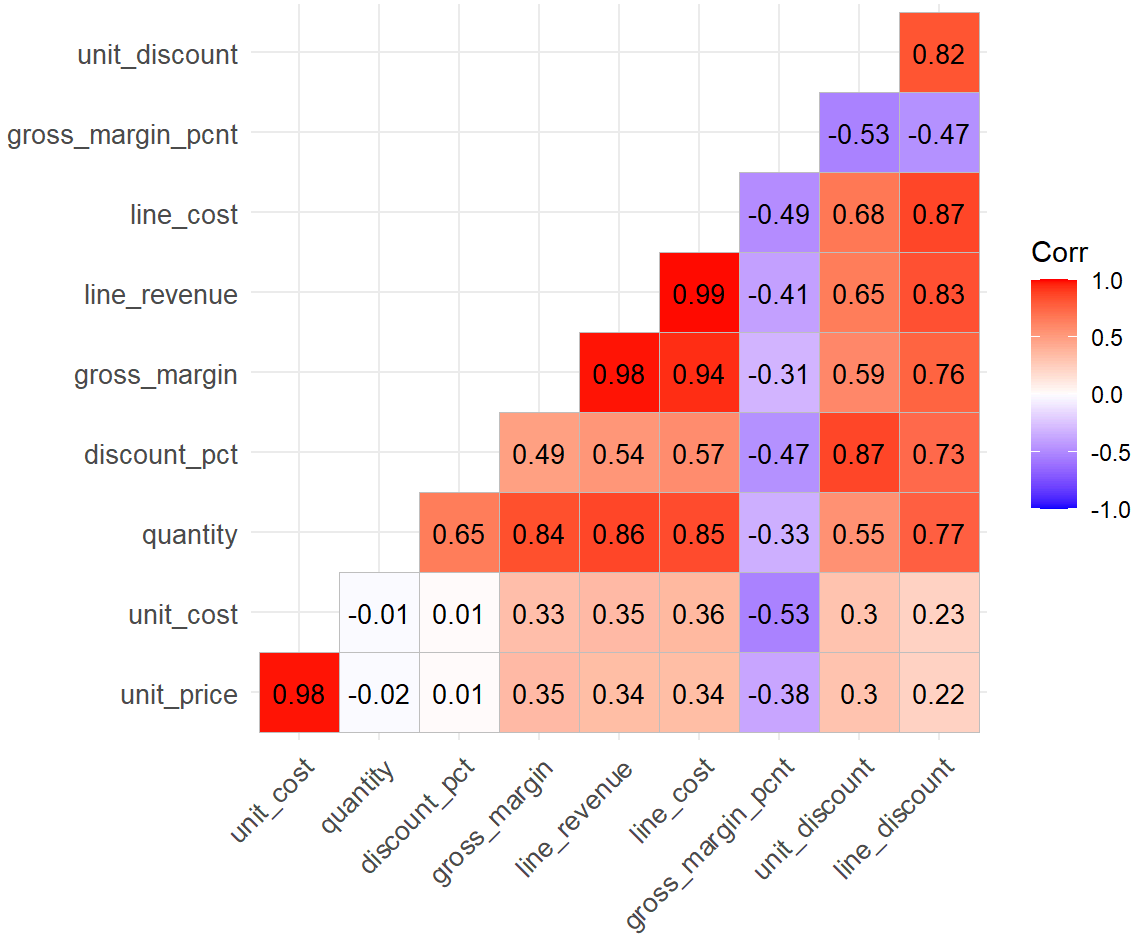
Evaluate boxplot by revenue by product category by loyalty customer type (New or Returning)



Findings:

- Across most cities, returning customers gave higher gross margin than new customers, which makes targeting the existing customer is ideal to increase the sales and profit. ### Correlation - Evaluate Relationship of Numeric Values

1. Evaluate relationship of all numeric values in a correlation matrix



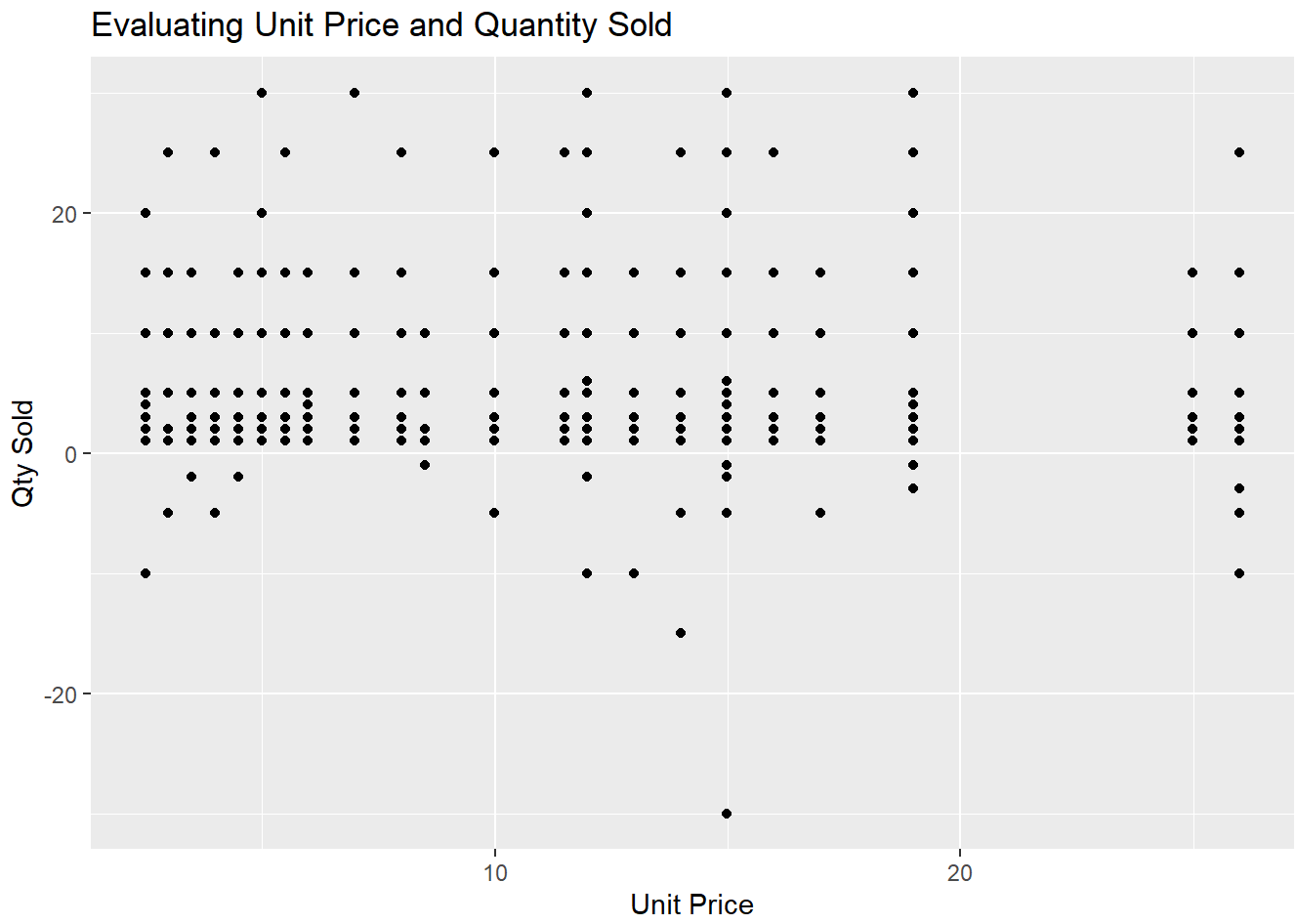
Strong positive Correlation:

- As quantity goes up gross margin, line revenue, line cost, and line discount goes up.
- Discount percent and unit discount are strongly correlated. As one goes up, other one goes up too.
- Gross margin goes up with Line revenue and line cost.
- Line revenue increases with increasing line cost and line discount
- Line cost and unit discount goes up with line discount,

Strong negative Correlation:

- If unit discount, line discount, line cost, and unit cost goes up gross margin percent goes down.

2. Evaluate Correlation of Key Numeric Values

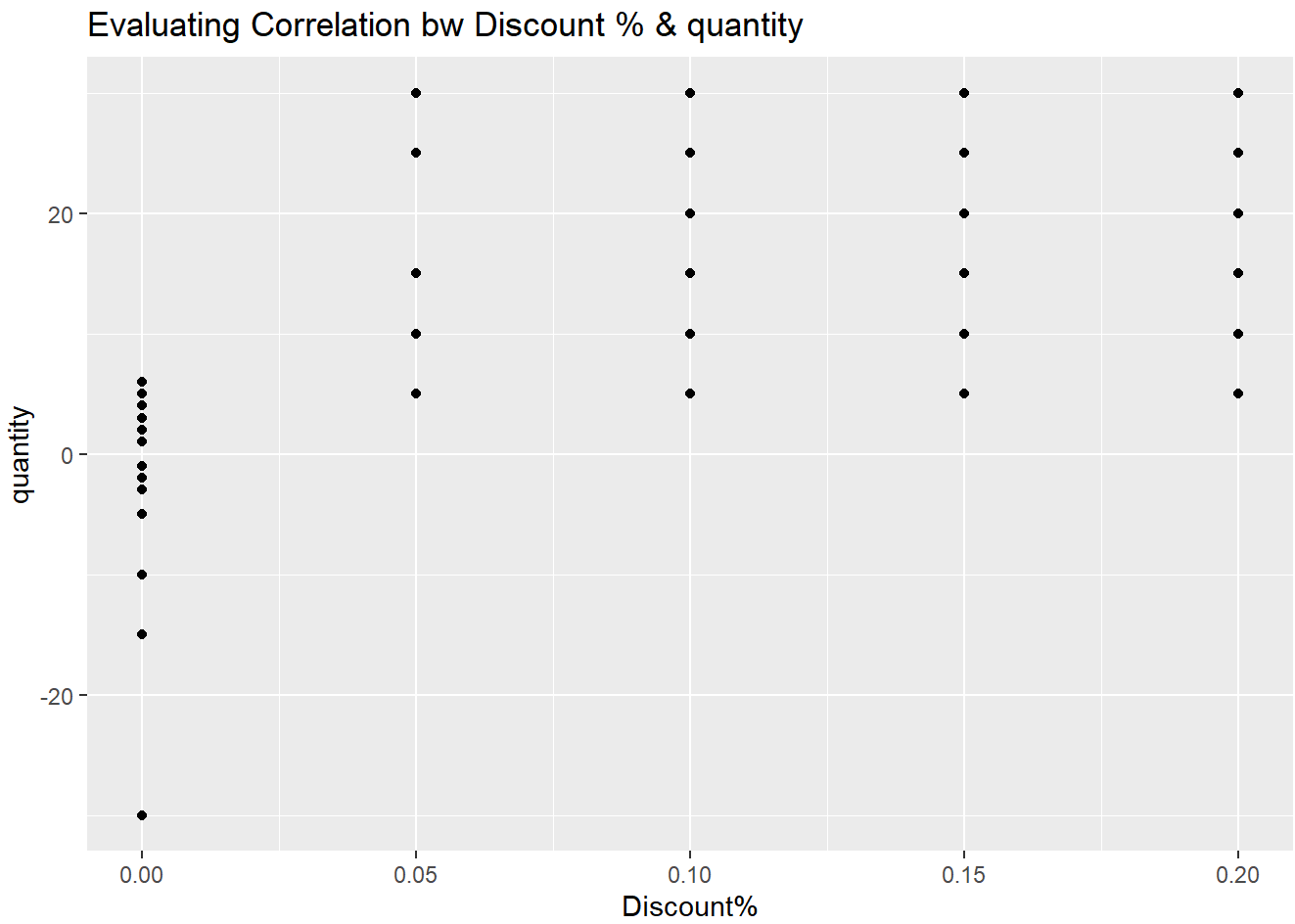



```
[1] -0.01875503
```

Finding:

- There is no strong correlation between unit price and quantity sold. The distribution is to scattered and there are some extreme values.

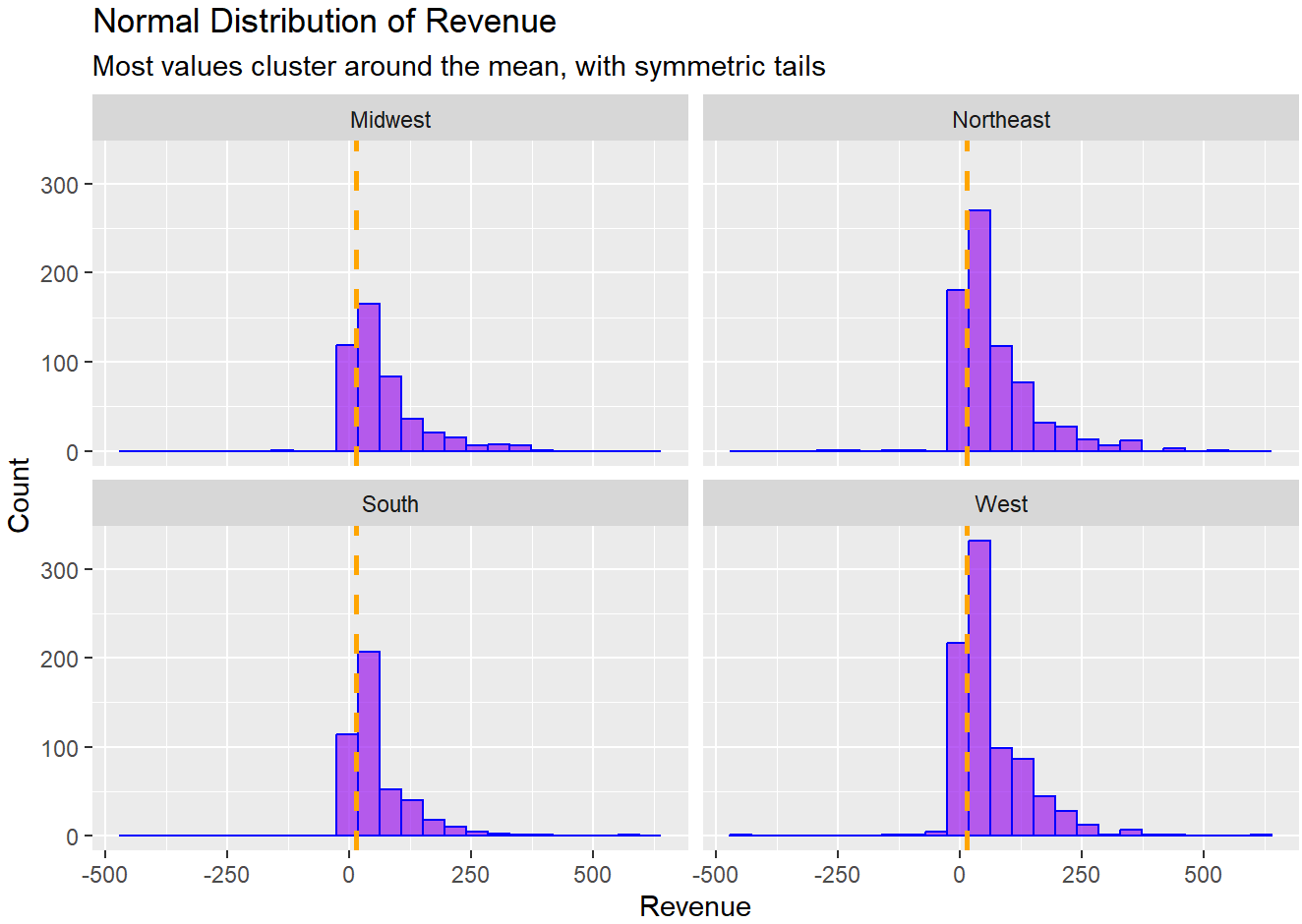
3. Evaluate correlation between discount level and quantity sold



Finding:

- There is no strong correlation between quantity and discount . The distribution is to scattered and there are some extreme values. ### Distributions - Evaluate Spread of Values

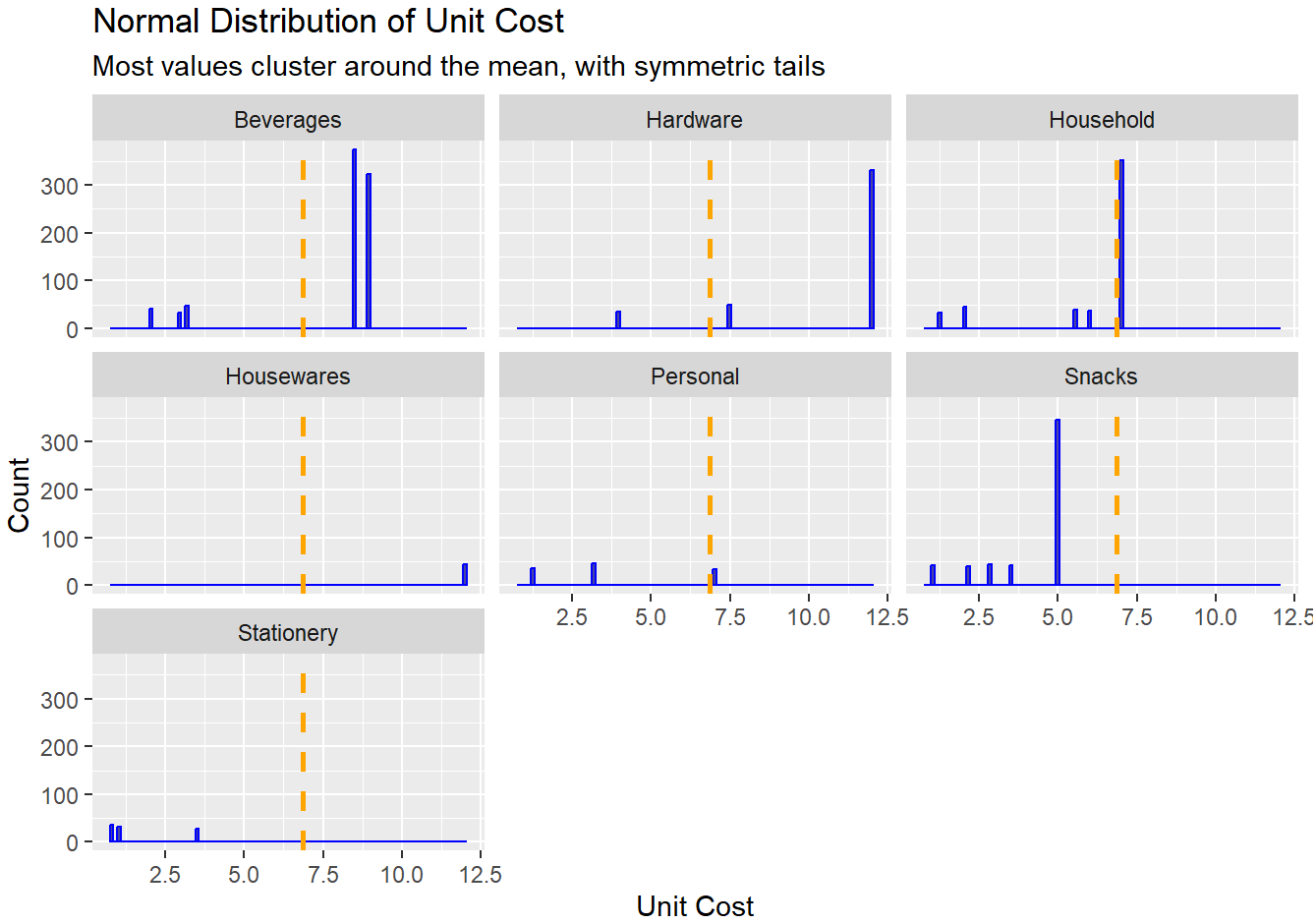
1. Evaluate Distribution of Revenue Data Points



Finding:

- All the graphs are skewed right and it means most of the low value small are in the left and on the right there are some high value product with low quantity.
- West and Northeast shows big spread.It shows more transaction and more profit.

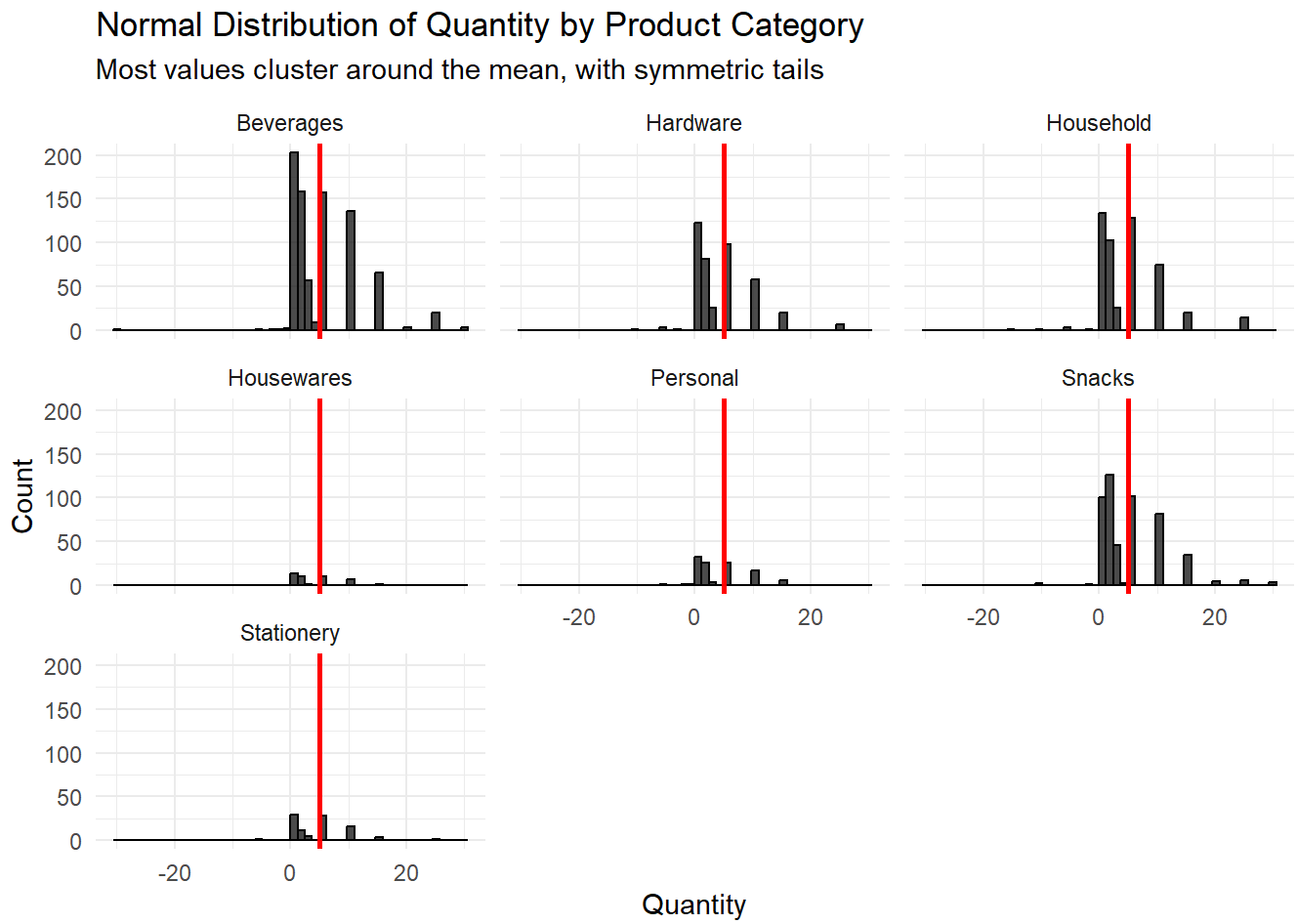
2. Evaluate Distribution of Unit Cost by Category



Findings:

- There are clusters on specific price point and non of the graph is bell shaped or normally distributed.
- Beverages, hardware, and Household has the highest peak, which shows higher cost values.
- The other product categories are clustered around lower price mark.

3. Evaluate Distribution of Quantity Sold By Product Category

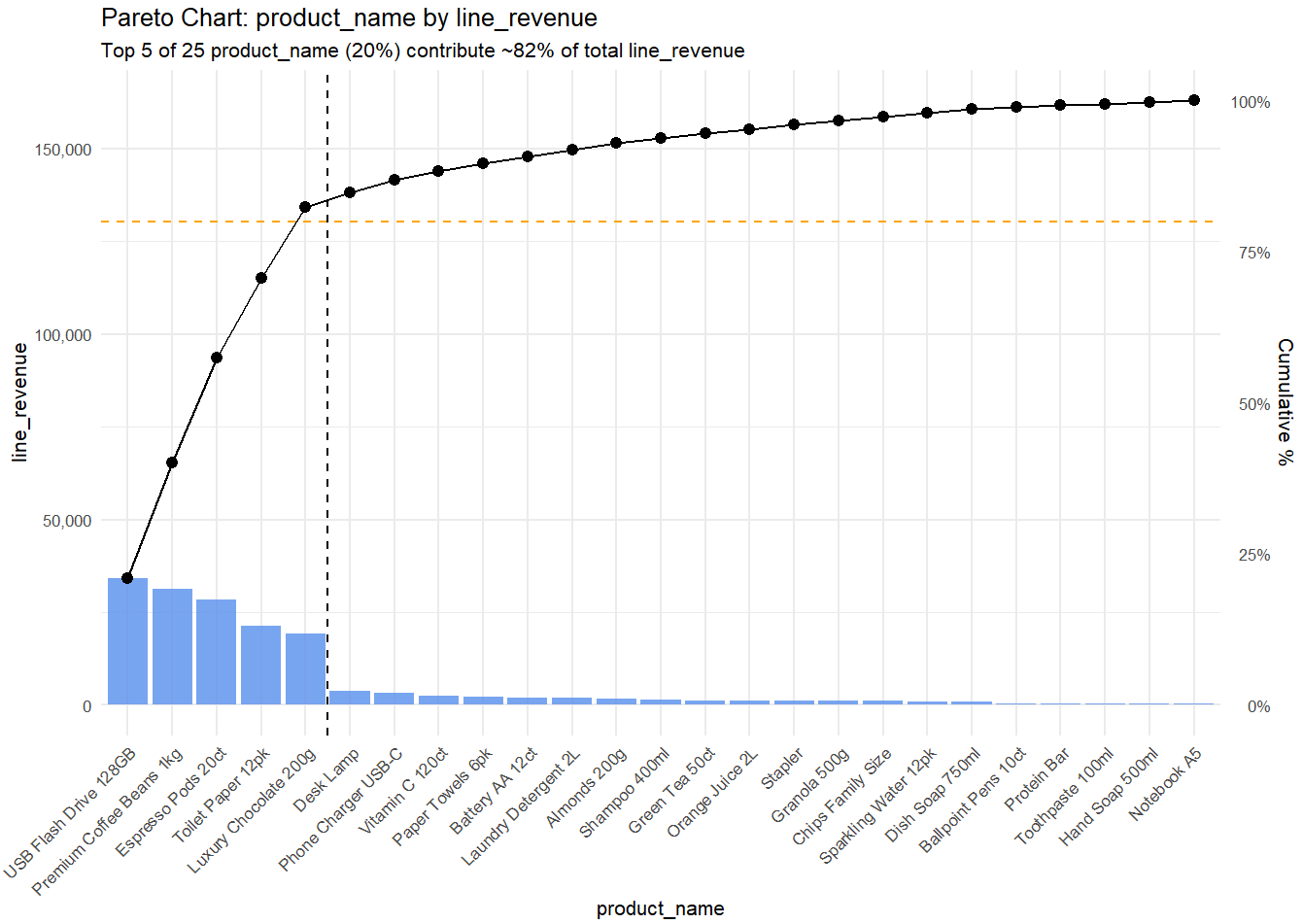


Findings:

- None of the graphs are normally distributed.
- Beverages, Hardware, Household, and snakes has some strong peaks, suggesting good performance in sales.
- Housewares, personal, and stationary are clustered on lower end. It means they are sold in small quantity.

Pareto - Identify Relationship of Inputs to Outputs ("80/20 Rule")

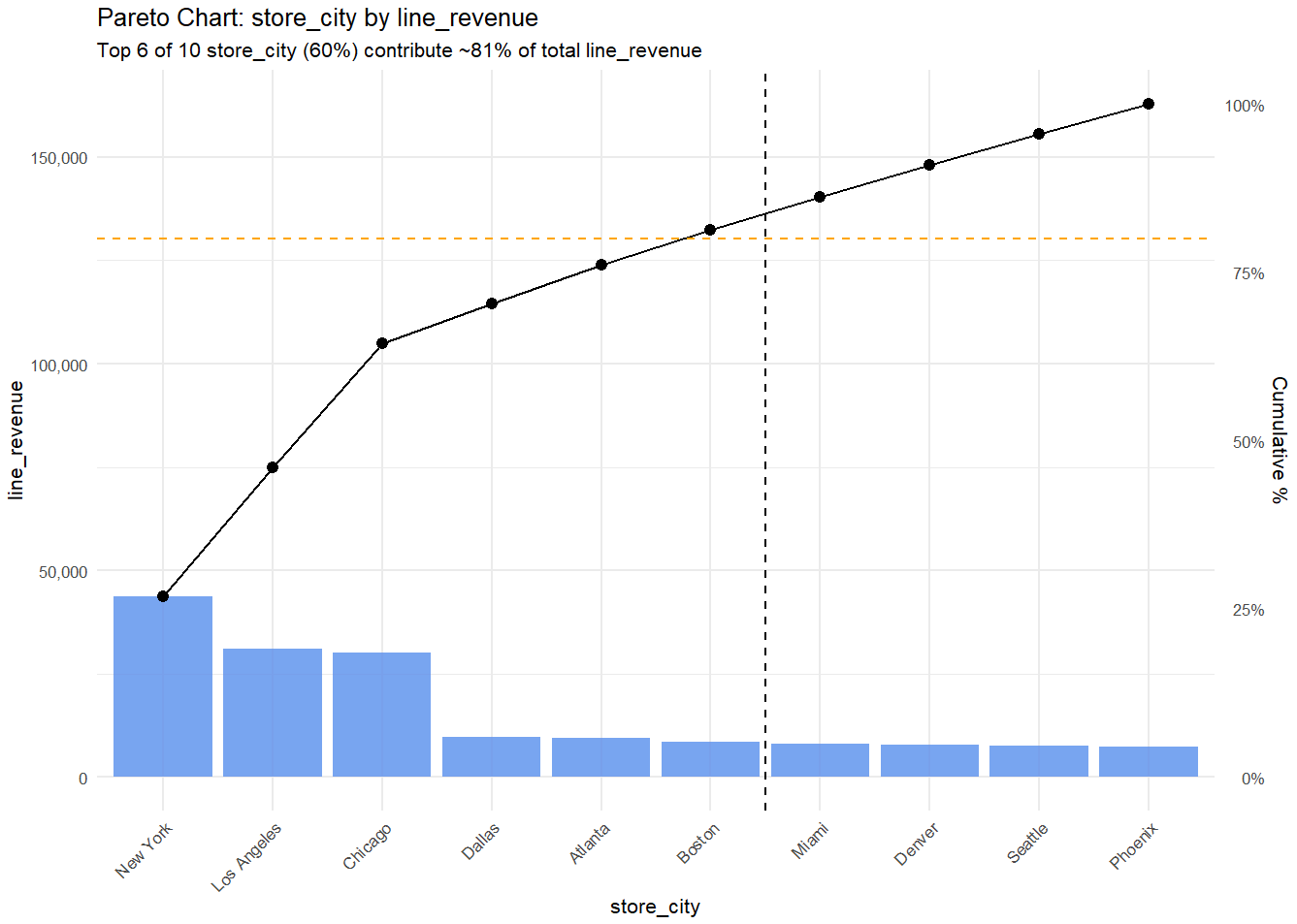
1. Perform a Pareto Analysis of Revenue by Product



Findings:

- Only 5 products accumulated about 80% of total revenue. These are only 20% of total products. They are USB Flash Drive 128GB, Premium Coffee Beans 1kg, Espresso Pods 20ct, Toilet Paper 12pk, Luxury Chocolate 200g.

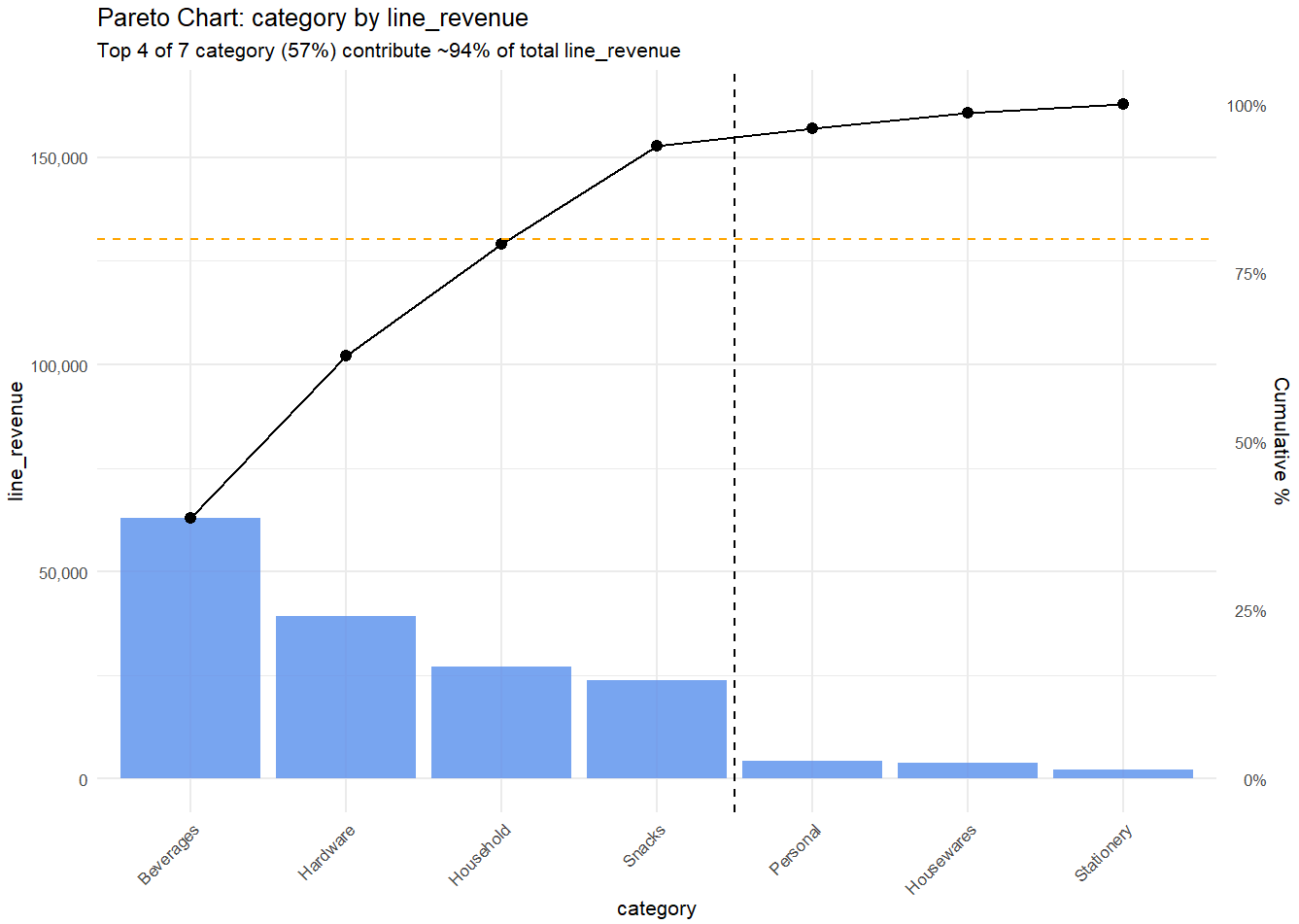
2. Perform a pareto analysis of revenue and store city location



Finding:

- Out of top 10 cities, 6 of them generated ~81% of total revenue. The 6 cities make 60% of the top cities. They are New York, Los Angeles, Chicago, Dallas, Atlanta, and Boston.

3. Perform a pareto analysis of revenue and Category location



Finding:

- Four products made ~94% of total revenue but are 57% of total products. They are Beverages, Hardware, Household, and snakes. ### Build A Report to Evaluate Impact of Promotions

1. Evaluate Impact of Promotions on Qty Sold, Unit Price, and Profit

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
Almonds 200g										
CLEARANCE20	3	40	$7.99	$255.68	$6.39	$115.68	$2.89	20.0%	$38.56	45.2%
NONE	21	44	$7.99	$351.56	$7.99	$197.56	$4.49	0.0%	$9.41	56.2%
SUMMER10	6	50	$7.99	$359.55	$7.19	$184.55	$3.69	10.0%	$30.76	51.3%
VIP15	7	80	$7.99	$543.32	$6.79	$263.32	$3.29	15.0%	$37.62	48.5%
WEEKEND5	5	40	$7.99	$303.62	$7.59	$163.62	$4.09	5.0%	$32.72	53.9%

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
Ballpoint Pens 10ct										
CLEARANCE20	1	5	$2.99	$11.96	$2.39	$7.96	$1.59	20.0%	$7.96	66.6%
NONE	14	21	$2.99	$62.79	$2.99	$45.99	$2.19	0.0%	$3.29	73.2%
SUMMER10	11	75	$2.99	$200.33	$2.67	$140.33	$1.87	9.1%	$12.76	70.5%
VIP15	7	55	$2.99	$139.78	$2.54	$95.78	$1.74	15.0%	$13.68	68.5%
WEEKEND5	3	30	$2.99	$85.22	$2.84	$61.22	$2.04	5.0%	$20.41	71.8%
Battery AA 12ct										
CLEARANCE20	1	25	$9.99	$199.80	$7.99	$99.80	$3.99	20.0%	$99.80	49.9%
NONE	17	27	$9.99	$269.73	$9.99	$161.73	$5.99	0.0%	$9.51	60.0%
SUMMER10	10	105	$9.99	$944.05	$8.99	$524.05	$4.99	10.0%	$52.41	55.5%
VIP15	3	40	$9.99	$339.66	$8.49	$179.66	$4.49	15.0%	$59.89	52.9%
WEEKEND5	3	10	$9.99	$92.41	$9.24	$52.41	$5.24	3.3%	$17.47	58.6%
Chips Family Size										
CLEARANCE20	2	10	$4.99	$39.92	$3.99	$17.92	$1.79	20.0%	$8.96	44.9%
NONE	20	41	$4.99	$204.59	$4.99	$114.39	$2.79	0.0%	$5.72	55.9%
SUMMER10	9	100	$4.99	$449.10	$4.49	$229.10	$2.29	10.0%	$25.46	51.0%
VIP15	6	85	$4.99	$360.53	$4.24	$173.53	$2.04	15.0%	$28.92	48.1%
WEEKEND5	2	10	$4.99	$47.41	$4.74	$25.41	$2.54	5.0%	$12.70	53.6%
Desk Lamp										
CLEARANCE20	5	45	$24.99	$899.64	$19.99	$359.64	$7.99	20.0%	$71.93	40.0%
NONE	25	37	$24.99	$924.63	$24.99	$480.63	$12.99	0.0%	$19.23	52.0%
SUMMER10	7	50	$24.99	$1,124.55	$22.49	$524.55	$10.49	10.0%	$74.94	46.6%
VIP15	2	15	$24.99	$318.62	$21.24	$138.62	$9.24	15.0%	$69.31	43.5%
WEEKEND5	4	25	$24.99	$593.51	$23.74	$293.51	$11.74	5.0%	$73.38	49.5%
Dish Soap 750ml										
CLEARANCE20	1	10	$4.49	$35.92	$3.59	$15.92	$1.59	20.0%	$15.92	44.3%
NONE	23	32	$4.49	$143.68	$4.49	$79.68	$2.49	0.0%	$3.46	55.5%
SUMMER10	16	145	$4.49	$585.95	$4.04	$295.95	$2.04	10.0%	$18.50	50.5%
VIP15	4	40	$4.49	$152.66	$3.82	$72.66	$1.82	15.0%	$18.16	47.6%

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
WEEKEND5	1	10	$4.49	$42.66	$4.27	$22.66	$2.27	5.0%	$22.66	53.1%
Espresso Pods 20ct										
CLEARANCE20	20	295	$14.99	$3,537.64	$11.99	$1,030.14	$3.49	20.0%	$51.51	29.1%
NONE	207	392	$14.99	$5,876.08	$14.99	$2,544.08	$6.49	0.0%	$12.29	43.3%
SUMMER10	69	660	$14.99	$8,904.06	$13.49	$3,294.06	$4.99	10.0%	$47.74	37.0%
VIP15	50	455	$14.99	$5,718.69	$12.57	$1,851.19	$4.07	14.4%	$37.02	33.7%
WEEKEND5	29	305	$14.99	$4,343.35	$14.24	$1,750.85	$5.74	5.0%	$60.37	40.3%
Granola 500g										
CLEARANCE20	3	20	$5.99	$95.84	$4.79	$39.84	$1.99	20.0%	$13.28	41.6%
NONE	25	48	$5.99	$287.52	$5.99	$153.12	$3.19	0.0%	$6.12	53.3%
SUMMER10	8	70	$5.99	$377.37	$5.39	$181.37	$2.59	10.0%	$22.67	48.1%
VIP15	4	45	$5.99	$229.12	$5.09	$103.12	$2.29	15.0%	$25.78	45.0%
WEEKEND5	3	20	$5.99	$113.81	$5.69	$57.81	$2.89	5.0%	$19.27	50.8%
Green Tea 50ct										
NONE	21	42	$7.99	$335.58	$7.99	$209.58	$4.99	0.0%	$9.98	62.5%
SUMMER10	5	60	$7.99	$431.46	$7.19	$251.46	$4.19	10.0%	$50.29	58.3%
VIP15	4	45	$7.99	$305.62	$6.79	$170.62	$3.79	15.0%	$42.65	55.8%
WEEKEND5	2	15	$7.99	$113.86	$7.59	$68.86	$4.59	5.0%	$34.43	60.5%
Hand Soap 500ml										
CLEARANCE20	1	5	$3.99	$15.96	$3.19	$9.96	$1.99	20.0%	$9.96	62.4%
NONE	21	39	$3.99	$155.61	$3.99	$108.81	$2.79	0.0%	$5.18	69.9%
SUMMER10	6	55	$3.99	$195.51	$3.55	$129.51	$2.35	8.3%	$21.59	67.1%
WEEKEND5	4	20	$3.99	$75.81	$3.79	$51.81	$2.59	5.0%	$12.95	68.3%
Laundry Detergent 2L										
CLEARANCE20	1	10	$12.99	$103.92	$10.39	$48.92	$4.89	20.0%	$48.92	47.1%
NONE	23	37	$12.99	$480.63	$12.99	$277.13	$7.49	0.0%	$12.05	57.7%
SUMMER10	6	40	$12.99	$467.64	$11.69	$247.64	$6.19	10.0%	$41.27	53.0%
VIP15	3	10	$12.99	$90.93	$9.09	$35.93	$3.59	10.0%	$11.98	52.7%
WEEKEND5	5	55	$12.99	$678.73	$12.34	$376.23	$6.84	5.0%	$75.25	55.4%

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
Luxury Chocolate 200g										
CLEARANCE20	11	80	$11.99	$767.36	$9.59	$367.36	$4.59	20.0%	$33.40	47.9%
NONE	203	382	$11.99	$4,580.18	$11.99	$2,670.18	$6.99	0.0%	$13.15	58.3%
SUMMER10	61	610	$11.99	$6,570.52	$10.77	$3,520.52	$5.77	9.8%	$57.71	53.7%
VIP15	37	340	$11.99	$3,465.11	$10.19	$1,765.11	$5.19	15.0%	$47.71	50.9%
WEEKEND5	34	335	$11.99	$3,815.82	$11.39	$2,140.82	$6.39	5.0%	$62.97	56.1%
Notebook A5										
CLEARANCE20	1	5	$3.49	$13.96	$2.79	$8.96	$1.79	20.0%	$8.96	64.2%
NONE	20	40	$3.49	$139.60	$3.49	$99.60	$2.49	0.0%	$4.98	71.3%
SUMMER10	4	25	$3.49	$78.53	$3.14	$53.53	$2.14	10.0%	$13.38	68.2%
VIP15	2	10	$3.49	$29.66	$2.97	$19.66	$1.97	15.0%	$9.83	66.3%
WEEKEND5	5	45	$3.49	$149.20	$3.32	$104.20	$2.32	5.0%	$20.84	69.8%
Orange Juice 2L										
NONE	35	55	$6.99	$384.45	$6.99	$208.45	$3.79	0.0%	$5.96	54.2%
SUMMER10	4	45	$6.99	$283.10	$6.29	$139.10	$3.09	10.0%	$34.77	49.1%
VIP15	4	35	$6.99	$207.95	$5.94	$95.95	$2.74	15.0%	$23.99	46.1%
WEEKEND5	4	45	$6.99	$298.82	$6.64	$154.82	$3.44	5.0%	$38.71	51.8%
Paper Towels 6pk										
CLEARANCE20	2	10	$11.49	$91.92	$9.19	$31.92	$3.19	20.0%	$15.96	34.7%
NONE	15	29	$11.49	$333.21	$11.49	$159.21	$5.49	0.0%	$10.61	47.8%
SUMMER10	8	90	$11.49	$930.69	$10.34	$390.69	$4.34	10.0%	$48.84	42.0%
VIP15	6	55	$11.49	$537.16	$9.77	$207.16	$3.77	15.0%	$34.53	38.6%
WEEKEND5	5	35	$11.49	$382.04	$10.92	$172.04	$4.92	5.0%	$34.41	45.0%
Phone Charger USB-C										
CLEARANCE20	2	40	$15.99	$511.68	$12.79	$211.68	$5.29	20.0%	$105.84	41.4%
NONE	29	65	$15.99	$1,039.35	$15.99	$551.85	$8.49	0.0%	$19.03	53.1%
SUMMER10	10	60	$15.99	$863.46	$14.39	$413.46	$6.89	10.0%	$41.35	47.9%
VIP15	4	20	$15.99	$271.83	$13.59	$121.83	$6.09	15.0%	$30.46	44.8%
WEEKEND5	5	40	$15.99	$607.62	$15.19	$307.62	$7.69	5.0%	$61.52	50.6%

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
Premium Coffee Beans 1kg										
CLEARANCE20	22	250	$18.99	$3,798.00	$15.19	$1,548.00	$6.19	20.0%	$70.36	40.8%
NONE	165	296	$18.99	$5,621.04	$18.99	$2,957.04	$9.99	0.0%	$17.92	52.6%
SUMMER10	63	555	$18.99	$9,485.50	$17.09	$4,490.50	$8.09	10.0%	$71.28	47.3%
VIP15	40	375	$18.99	$6,053.06	$16.14	$2,678.06	$7.14	15.0%	$66.95	44.2%
WEEKEND5	34	345	$18.99	$6,223.97	$18.04	$3,118.97	$9.04	5.0%	$91.73	50.1%
Protein Bar										
CLEARANCE20	2	20	$2.49	$39.84	$1.99	$19.84	$0.99	20.0%	$9.92	49.8%
NONE	20	48	$2.49	$119.52	$2.49	$71.52	$1.49	0.0%	$3.58	59.8%
SUMMER10	8	60	$2.49	$134.46	$2.24	$74.46	$1.24	10.0%	$9.31	55.4%
VIP15	8	70	$2.49	$144.42	$2.06	$74.42	$1.06	13.1%	$9.30	53.6%
WEEKEND5	3	25	$2.49	$59.14	$2.37	$34.14	$1.37	5.0%	$11.38	57.7%
Shampoo 400ml										
CLEARANCE20	1	10	$8.49	$67.92	$6.79	$35.92	$3.59	20.0%	$35.92	52.9%
NONE	26	33	$8.49	$280.17	$8.49	$174.57	$5.29	0.0%	$6.71	62.3%
SUMMER10	7	55	$8.49	$420.25	$7.64	$244.25	$4.44	10.0%	$34.89	58.1%
VIP15	7	55	$8.49	$396.91	$7.22	$220.91	$4.02	15.0%	$31.56	55.7%
WEEKEND5	4	25	$8.49	$201.64	$8.07	$121.64	$4.87	5.0%	$30.41	60.3%
Sparkling Water 12pk										
CLEARANCE20	3	30	$5.49	$131.76	$4.39	$71.76	$2.39	20.0%	$23.92	54.5%
NONE	26	51	$5.49	$279.99	$5.49	$177.99	$3.49	0.0%	$6.85	63.6%
SUMMER10	7	80	$5.49	$395.28	$4.94	$235.28	$2.94	10.0%	$33.61	59.5%
VIP15	3	35	$5.49	$163.33	$4.67	$93.33	$2.67	15.0%	$31.11	57.1%
WEEKEND5	1	5	$5.49	$26.08	$5.22	$16.08	$3.22	5.0%	$16.08	61.7%
Stapler										
CLEARANCE20	4	55	$7.99	$351.56	$6.39	$159.06	$2.89	20.0%	$39.77	45.2%
NONE	14	20	$7.99	$159.80	$7.99	$89.80	$4.49	0.0%	$6.41	56.2%
SUMMER10	7	60	$7.99	$431.46	$7.19	$221.46	$3.69	10.0%	$31.64	51.3%
WEEKEND5	3	25	$7.99	$189.76	$7.59	$102.26	$4.09	5.0%	$34.09	53.9%

Promo Performance by Product

Sales, Revenue, Profitability, and Discounts

	Transactions	Total Qty Sold	Avg Unit Price	Total Revenue	Revenue / Unit	Total Profit	Profit / Unit	Avg Discount %	Avg Gross Margin	Avg GM %
Toilet Paper 12pk										
CLEARANCE20	14	95	$13.99	$1,049.25	$11.04	$384.25	$4.04	18.6%	$27.45	38.3%
NONE	199	369	$13.99	$5,162.31	$13.99	$2,579.31	$6.99	0.0%	$12.96	50.0%
SUMMER10	77	625	$13.99	$7,841.40	$12.55	$3,466.40	$5.55	9.7%	$45.02	44.5%
VIP15	35	295	$13.99	$3,507.99	$11.89	$1,442.99	$4.89	15.0%	$41.23	41.1%
WEEKEND5	27	290	$13.99	$3,854.24	$13.29	$1,824.24	$6.29	5.0%	$67.56	47.3%
Toothpaste 100ml										
CLEARANCE20	1	15	$3.49	$41.88	$2.79	$23.88	$1.59	20.0%	$23.88	57.0%
NONE	21	30	$3.49	$104.70	$3.49	$68.70	$2.29	0.0%	$3.27	65.6%
SUMMER10	7	45	$3.49	$141.35	$3.14	$87.35	$1.94	10.0%	$12.48	61.8%
VIP15	4	35	$3.49	$103.83	$2.97	$61.83	$1.77	15.0%	$15.46	59.5%
WEEKEND5	2	20	$3.49	$66.31	$3.32	$42.31	$2.12	5.0%	$21.16	63.8%
USB Flash Drive 128GB										
CLEARANCE20	18	130	$25.99	$2,676.97	$20.59	$1,116.97	$8.59	18.9%	$62.05	42.9%
NONE	197	334	$25.99	$8,680.66	$25.99	$4,672.66	$13.99	0.0%	$23.72	53.8%
SUMMER10	51	410	$25.99	$9,577.31	$23.36	$4,657.31	$11.36	9.8%	$91.32	48.8%
VIP15	37	345	$25.99	$7,621.57	$22.09	$3,481.57	$10.09	15.0%	$94.10	45.7%
WEEKEND5	29	225	$25.99	$5,542.37	$24.63	$2,842.37	$12.63	4.8%	$98.01	51.5%
Vitamin C 120ct										
NONE	16	27	$16.99	$458.73	$16.99	$269.73	$9.99	0.0%	$16.86	58.8%
SUMMER10	8	80	$16.99	$1,223.28	$15.29	$663.28	$8.29	10.0%	$82.91	54.2%
VIP15	5	15	$16.99	$203.88	$13.59	$98.88	$6.59	12.0%	$19.78	53.0%
WEEKEND5	4	30	$16.99	$484.22	$16.14	$274.22	$9.14	5.0%	$68.55	56.6%

Impact of Quantity sold:

- Promotions significantly increased quantity sold. All the promo code showed high sales in numbers.

Impact of Unit price:

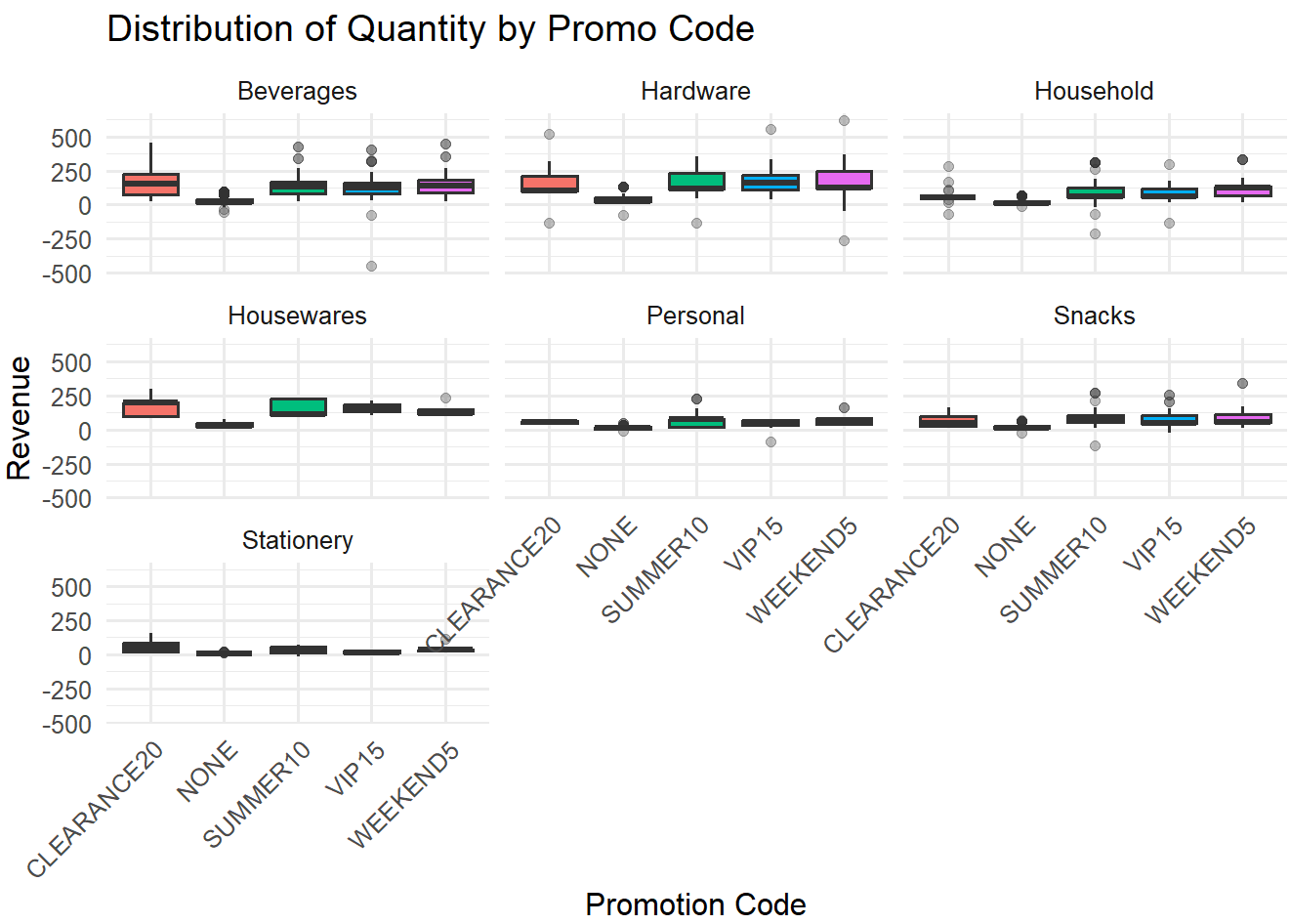
- Promos reduces the prices on product but the higher the number of sales.

Impact on Profit:

- Even though unit price decreases because of promo codes which lowers the profit too but the boost in sales because of promo increases the profit than in reduces.

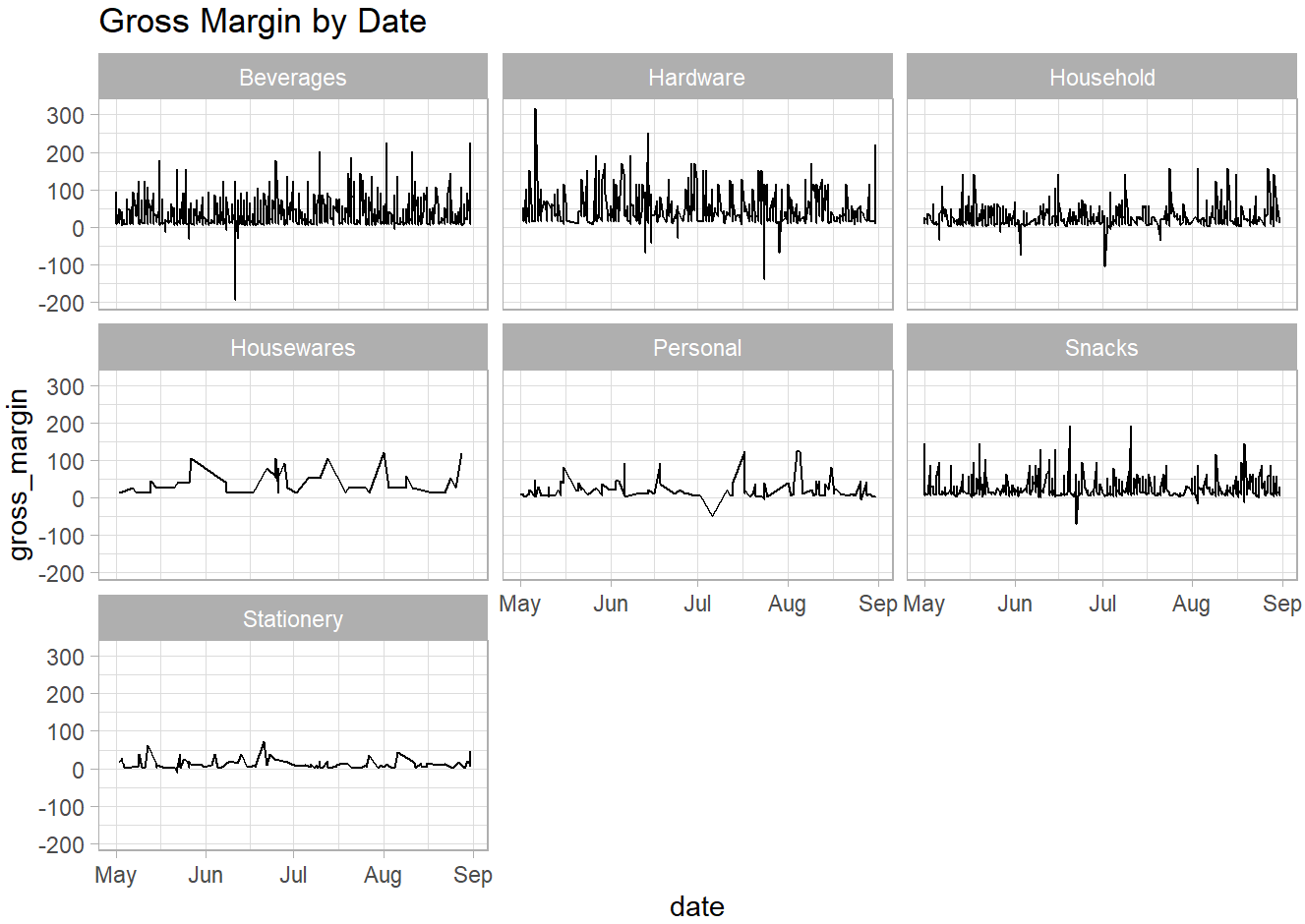
2. Evaluate Relationship of Promotions to Quantity Sold For All Products - Visually

Visualizing complex relationships is a powerful way to derive meaning from data.

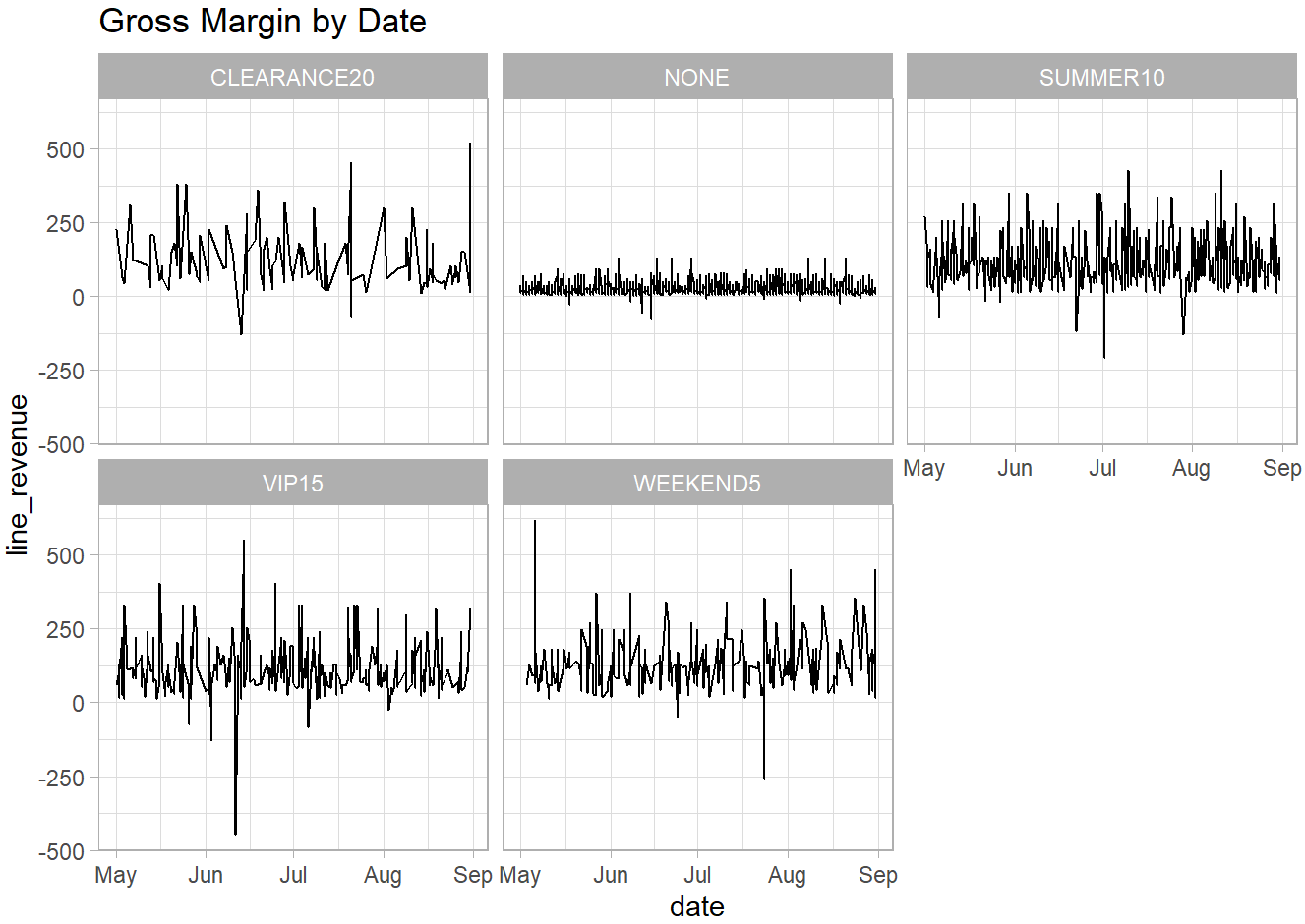


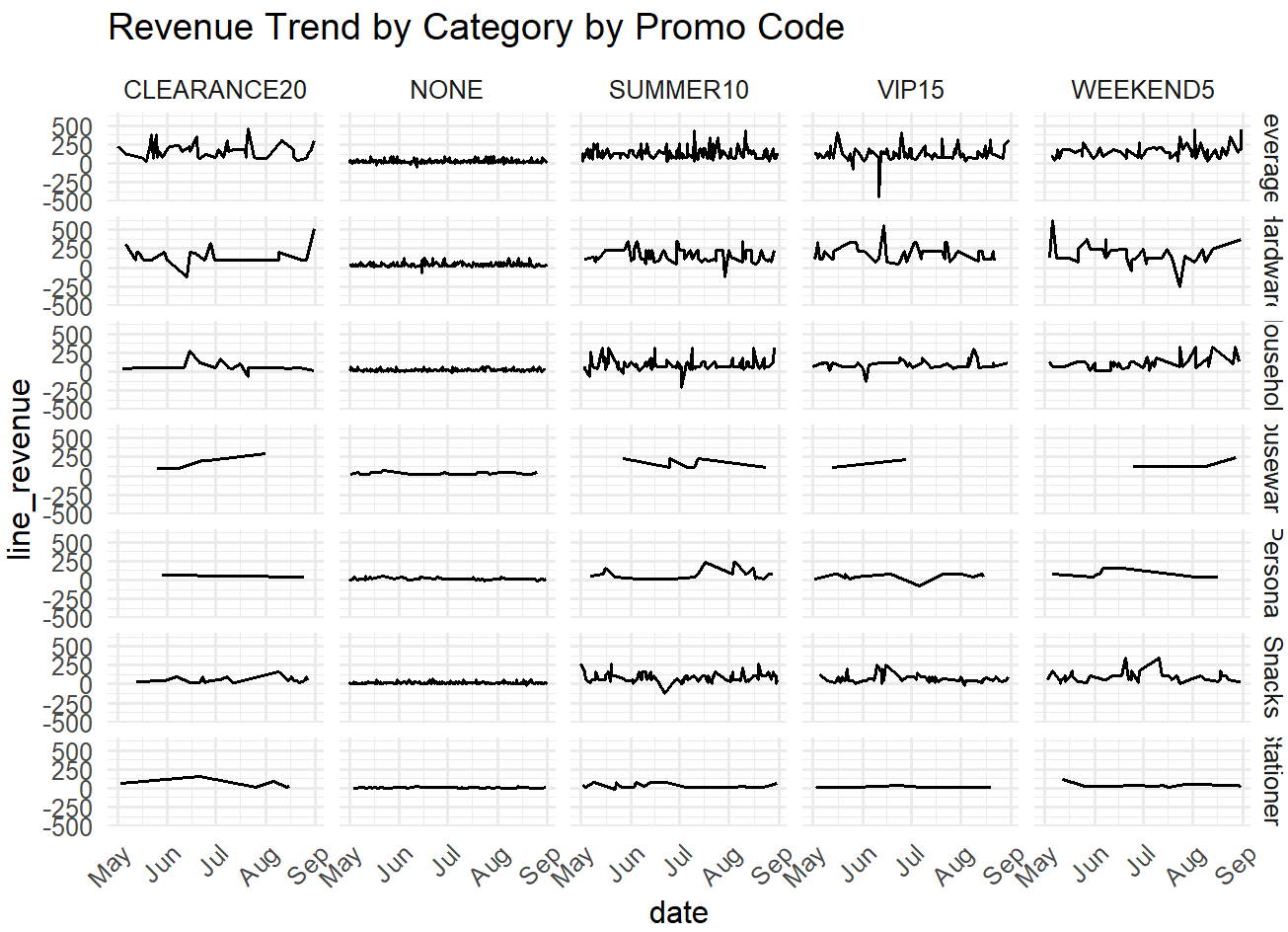
Finding:

- Beverage, household, housewares, and hardware are greatly impacted by promo codes. These promo codes helped sales in more higher quantity.
- Among the promos SUMMER10, CLEARANCE are effective and after these VIP15 is good.
-



3. Now evaluate revenue by Promotion Code Over Time

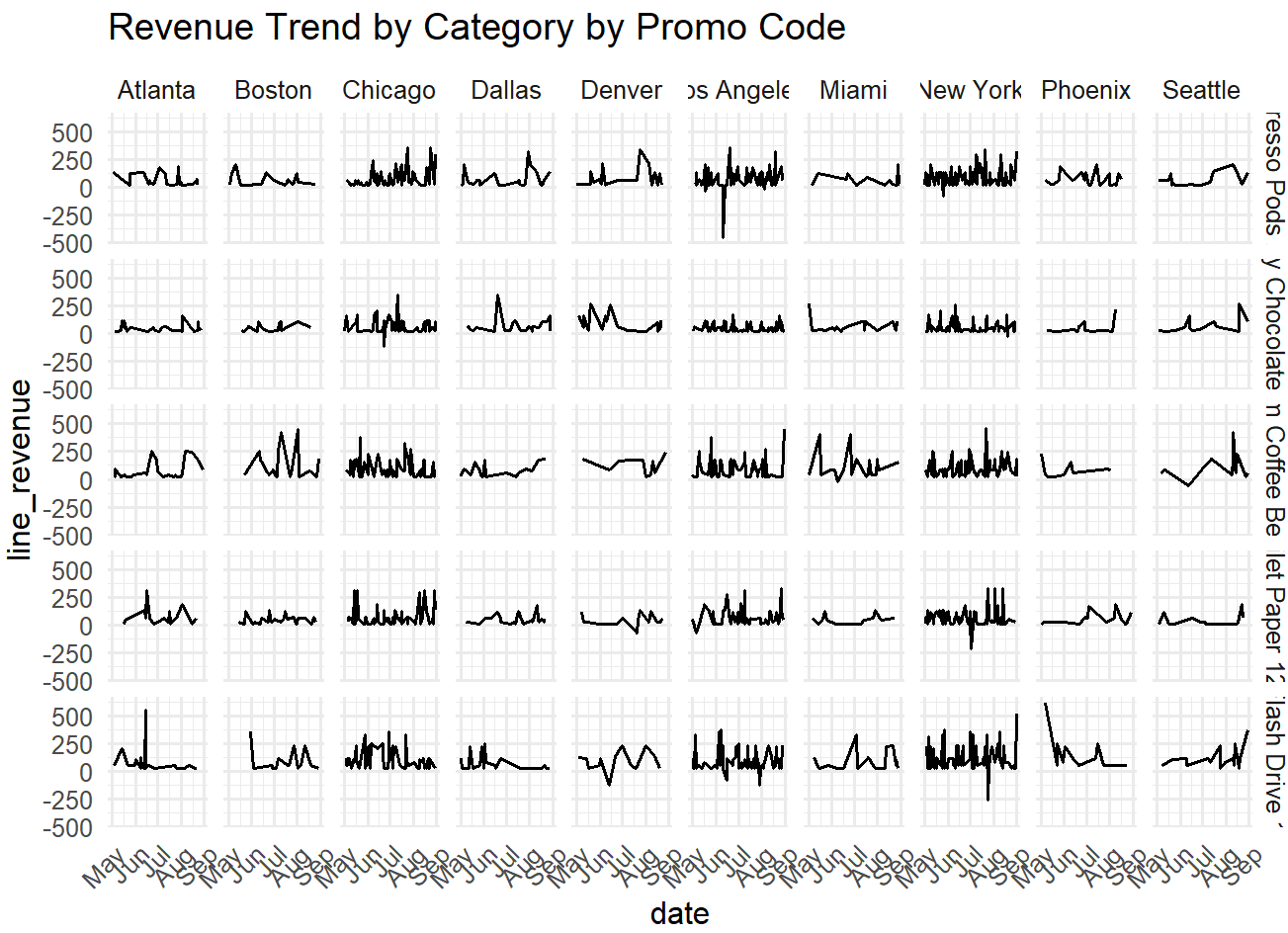

Gross Margin by Date
CLEARANCE20
NONE
SUMMER10
VIP15
WEEKEND5
500
250
0
-250
-500
line_revenue
May
Jun
Jul
Aug
Sep
date



Findings:

- SUMMER10 and VIP15 shows that they have generated more revenue among all the products.
- CLEARANCE also showed strong revenue but mostly with the beverages.
- WEEKEND5 shows more strong and consistent revenue across all category.
- NONE is mostly flat line among all the categories.

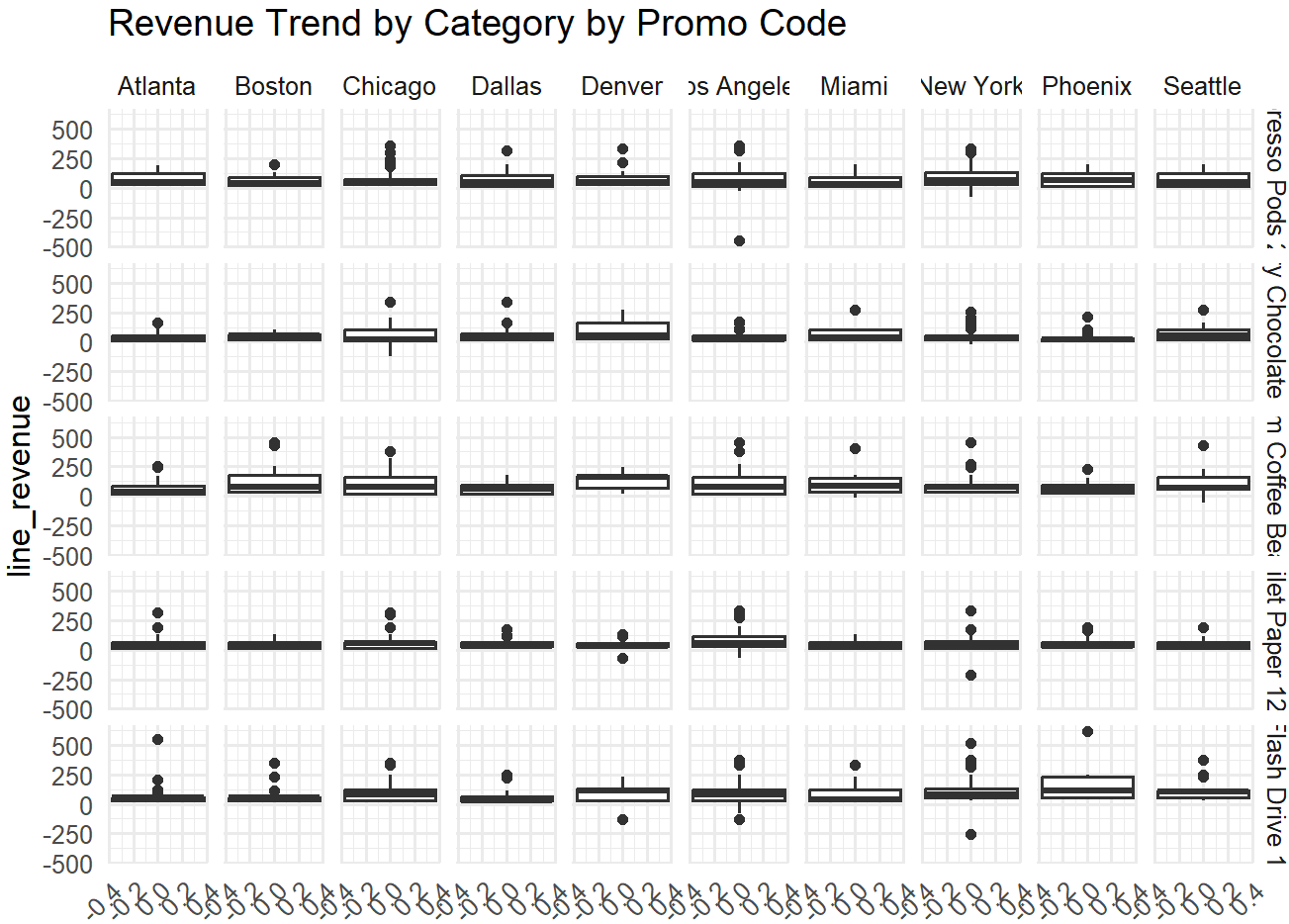
4. Now evaluate Revenue by Product by City Over Time



Finding:

- New York, Chicago, and Los Angeles shows the highest number of revenue spikes across products.
- Dallas, Denver, and Seattle shows moderate revenue spike.
- Boston and Phoenix are mostly flatter revenue but occasionally a little spike.

5. Now evaluate revenue by Product by City



Analyze Data

The exploration phase of the analytic process is intended to expansive and generative, asking "what does the data tell me?" to the analysis phase, which is more focused on **outcome** in relation to the objective, and asks "what does the data mean?".

It is a subtle but important mind shift.

Analysis 1: Analyze Sales Data

Question: What was overall sales revenue across all stores?

```
[1] Phoenix
10 Levels: Atlanta Boston Chicago Dallas Denver Los Angeles Miami ... Seattle
```

```
[1] ""
```

Our total sales revenue was **$162,888**.

Question: What are the best selling products?

```
[1] "$34,098.88"
```

```
# A tibble: 1 × 1
  product_name
  <fct>
1 USB Flash Drive 128GB
```

The best selling product is $34,098.88 and brought in **0**

Display a nice table of the best selling 15 products:

product_name	TOTAL_REVENUE
USB Flash Drive 128GB	$34,098.88
Premium Coffee Beans 1kg	$31,181.58
Espresso Pods 20ct	$28,379.82
Toilet Paper 12pk	$21,415.19
Luxury Chocolate 200g	$19,198.99
Desk Lamp	$3,860.96
Phone Charger USB-C	$3,293.94
Vitamin C 120ct	$2,370.10
Paper Towels 6pk	$2,275.02
Battery AA 12ct	$1,845.65
Laundry Detergent 2L	$1,821.85
Almonds 200g	$1,813.73
Shampoo 400ml	$1,366.89
Green Tea 50ct	$1,186.52
Orange Juice 2L	$1,174.32

Analysis 2: Analyze Profit Data

Question: What is the overall profitability (Revenue - Cost)?

```
[1] "$77,135.06"
```

Our total profit was **$77,135.06**

Question: Which Products are the most profitable?

```
[1] Ballpoint Pens 10ct  Notebook A5          Hand Soap 500ml
[4] Toothpaste 100ml     Sparkling Water 12pk
25 Levels: Almonds 200g Ballpoint Pens 10ct ... Vitamin C 120ct
```

The most profitable product is Ballpoint Pens 10ct, Notebook A5, Hand Soap 500ml, Toothpaste 100ml, Sparkling Water 12pk and the total profit is 0

Here are the top 15 most profitable items: 1.Sparkling Water 12pk
2.Dish Soap 750ml
3.Ballpoint Pens 10ct
4.Protein Bar
5.Toothpaste 100ml
6.Hand Soap 500ml
7.Notebook A5
8.USB Flash Drive 128GB 9.Premium Coffee Beans 1kg
10.Desk Lamp 11.Phone Charger USB-C
12.Espresso Pods 20ct
13.Toilet Paper 12pk
14.Vitamin C 120ct
15.Paper Towels 6pk

```
# A tibble: 15 × 2
   product_name             TOTAL_REVENUE
   <fct>                    <chr>
 1 Sparkling Water 12pk     $996.44
 2 Dish Soap 750ml          $960.86
 3 Ballpoint Pens 10ct      $500.08
 4 Protein Bar              $497.38
 5 Toothpaste 100ml         $458.06
 6 Hand Soap 500ml          $442.89
 7 Notebook A5              $410.95
 8 USB Flash Drive 128GB    $34,098.88
 9 Premium Coffee Beans 1kg $31,181.58
10 Desk Lamp                $3,860.96
11 Phone Charger USB-C      $3,293.94
12 Espresso Pods 20ct       $28,379.82
13 Toilet Paper 12pk        $21,415.19
14 Vitamin C 120ct          $2,370.10
15 Paper Towels 6pk         $2,275.02
```

Analysis 3: Analyze Location Data

Question: What locations generated the most revenue? 1.New York
2.Los Angeles
3.Chicago
4.Dallas

5.Atlanta
6.Boston
7.Miami 8.Denver
9.Seattle
10.Phoenix

```
# A tibble: 10 × 2
   store_city  total_revenue
   <fct>               <dbl>
 1 New York           43772.
 2 Los Angeles        31084.
 3 Chicago            30113.
 4 Dallas              9580.
 5 Atlanta             9359.
 6 Boston              8428.
 7 Miami               7973.
 8 Denver              7857.
 9 Seattle             7471.
10 Phoenix             7251.
```

Question: What locations generated the most profit?

1.New York 2.Los Angeles 3.Chicago 4.Dallas 5.Atlanta

```
[1] New York    Los Angeles Chicago     Dallas      Atlanta
10 Levels: Atlanta Boston Chicago Dallas Denver Los Angeles Miami ... Seattle
```

Analysis 4: Analyze Customer Type & Loyalty Data

Question: What customer types generated the most revenue?

```
# A tibble: 20 × 12
# Groups:   customer_type, loyalty_member [4]
   customer_type loyalty_member promo_code  total_revenue total_quantity
   <fct>         <fct>          <fct>               <dbl>          <int>
 1 New           False          CLEARANCE20         4529.            385
 2 New           False          NONE               12691.            852
 3 New           False          SUMMER10           19791.           1590
 4 New           False          VIP15              11701.            925
 5 New           False          WEEKEND5            8884.            640
 6 New           True           CLEARANCE20         1471.            105
 7 New           True           NONE                3866.            263
 8 New           True           SUMMER10            3777.            315
 9 New           True           VIP15               3179.            230
10 New           True           WEEKEND5            2483.            180
11 Returning     False          CLEARANCE20         2870.            210
12 Returning     False          NONE                4645.            316
13 Returning     False          SUMMER10            8793.            735
14 Returning     False          VIP15               4519.            390
15 Returning     False          WEEKEND5            4554.            345
```

```
16 Returning     True           CLEARANCE20         5868.            505
17 Returning     True           NONE               15234.           1108
18 Returning     True           SUMMER10           20055.           1570
19 Returning     True           VIP15              11507.           1010
20 Returning     True           WEEKEND5           12471.            865
# ℹ 7 more variables: total_profit <dbl>, revenue_per_unit <dbl>,
#   avg_unit_price <dbl>, total_discount <dbl>, disc_unit <dbl>,
#   mean_gm_pcnt <dbl>, gm_per_unit <dbl>

# A tibble: 1 × 10
  customer_type total_revenue total_quantity total_profit revenue_per_unit
  <fct>                 <dbl>          <int>        <dbl>            <dbl>
1 Returning            90517.           7054       43052.             12.8
# ℹ 5 more variables: avg_unit_price <dbl>, total_discount <dbl>,
#   disc_unit <dbl>, mean_gm_pcnt <dbl>, gm_per_unit <dbl>
```

The customer type that generated more revenue is returning.

Question: What customer type generated the most profit?

```
[1] Returning
Levels: New Returning
```

Most revenue was generated through Returning customer.

Question: Were Loyalty Programs Effective? Look at Revenue & Profit

Analysis 5: Analyze Product Data

Question: What products generated the most revenue?

```
[1] USB Flash Drive 128GB
25 Levels: Almonds 200g Ballpoint Pens 10ct ... Vitamin C 120ct
```

USB Flash Drive 128GB

Question: What products generated the most profit?

```
[1] USB Flash Drive 128GB    Premium Coffee Beans 1kg Espresso Pods 20ct
[4] Toilet Paper 12pk        Luxury Chocolate 200g
25 Levels: Almonds 200g Ballpoint Pens 10ct ... Vitamin C 120ct
```

Top 5 products: 1. USB Flash Drive 128GB 2. Premium Coffee Beans 1kg 3. Espresso Pods 20ct 4. Toilet Paper 12pk 5. Luxury Chocolate 200g

Question: What product is the highest REVENUE generating overall?

```
# A tibble: 1 × 2
  product_name          total_revenue
  <fct>                         <dbl>
1 USB Flash Drive 128GB        34099.
```

Highest revenue generating product is USB Flash Drive 128GB with total revenue of 34098.88.

Question: What product is the highest PROFIT (gross margin) generating overall?

```
# A tibble: 1 × 2
  product_name          total_gross_margin
  <fct>                              <dbl>
1 USB Flash Drive 128GB             16771.
```

USB Flash Drive 128GB is most profitable product.

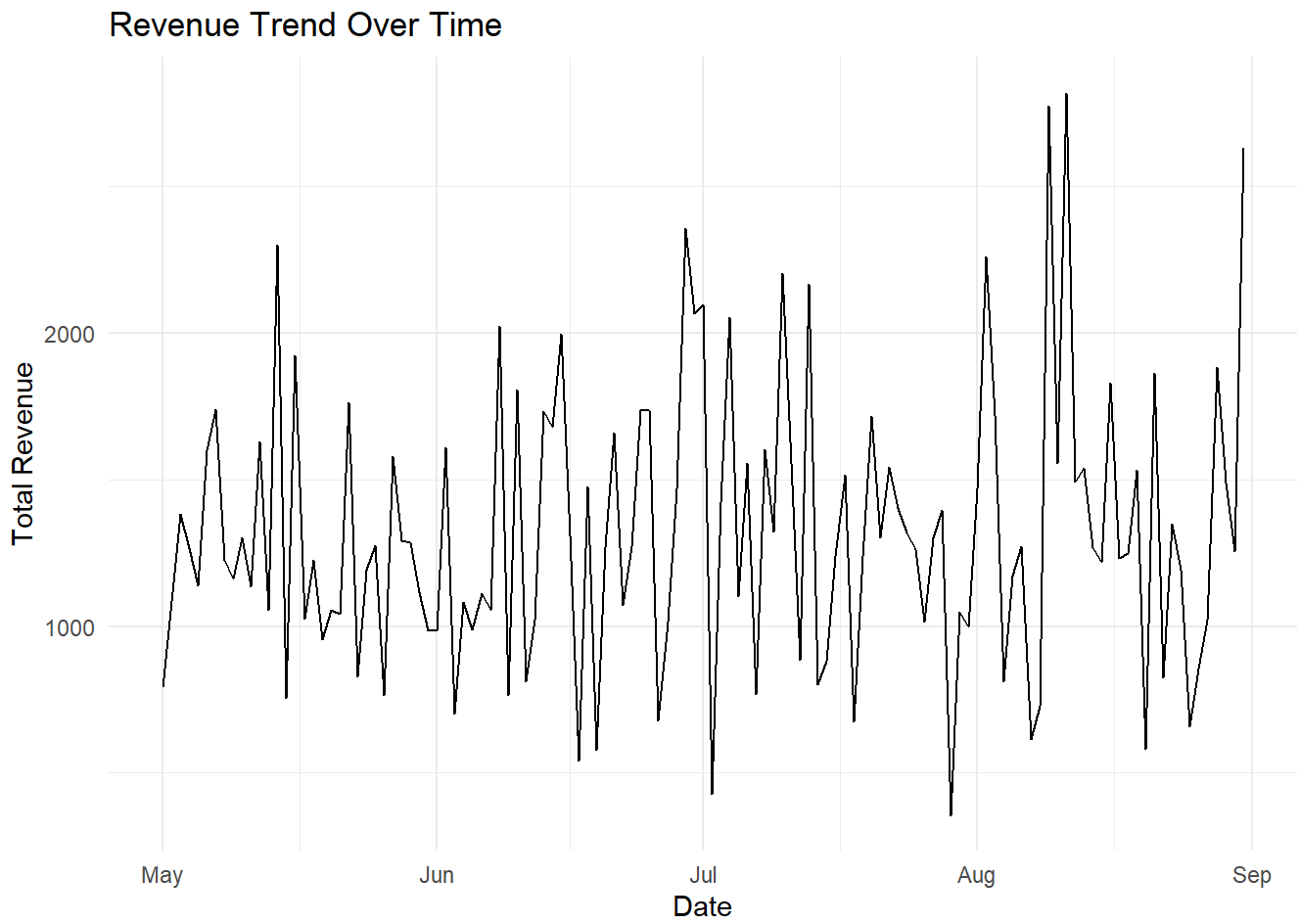
Question: Were the promotions effective? If so, which were effective?

```
# A tibble: 5 × 4
  promo_code  total_revenue total_gross_margin avg_gm_percent
  <fct>               <dbl>              <dbl>          <dbl>
1 SUMMER10           52416.             24660.          0.487
2 NONE               36436.             19123.          0.538
3 WEEKEND5           28392.             14180.          0.517
4 VIP15              30906.             13446.          0.454
5 CLEARANCE20        14738.              5725.          0.412
```

The best performing promo code was SUMMER10 by gross margin but by gross percentage is was NONE.

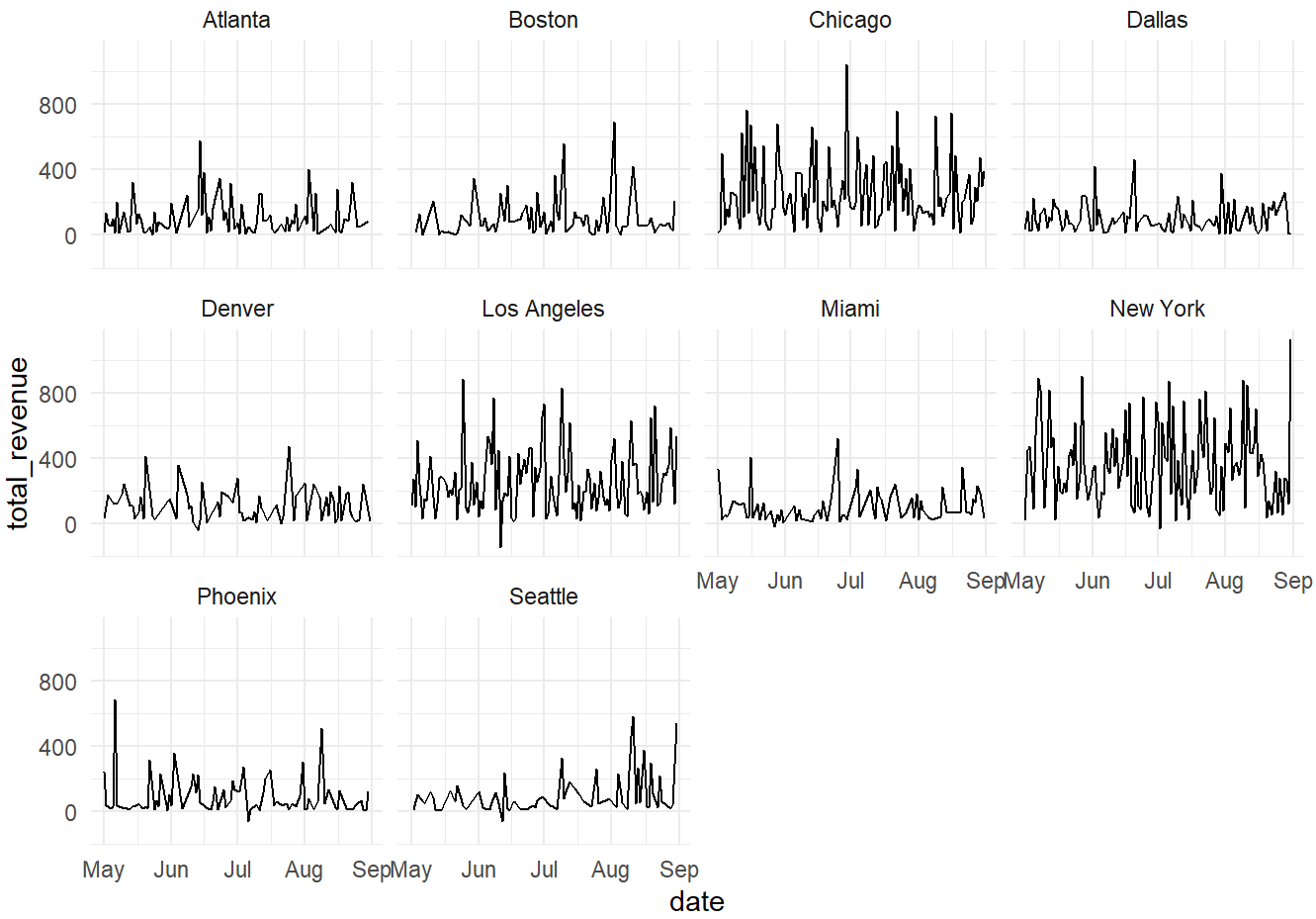
Analysis 6: Evaluate the Revenue Trend Over the Year (aggregate sales)

Question: Is sales revenue trending UP or DOWN?



The trend of sales revenue is mixed. Overall trend is upwards but if analyzed with details, it can be seen that May and June had low trend, between late June and late August there was a upward trend. After late August to september there is a sharper upward trend.

Question: How is sales revenue trending UP or DOWN by Location?



Upward trend:

New York- Strong upward trend with higher peaks around August-september.

Chicago- Similar increasing trend over the months with high peaks.

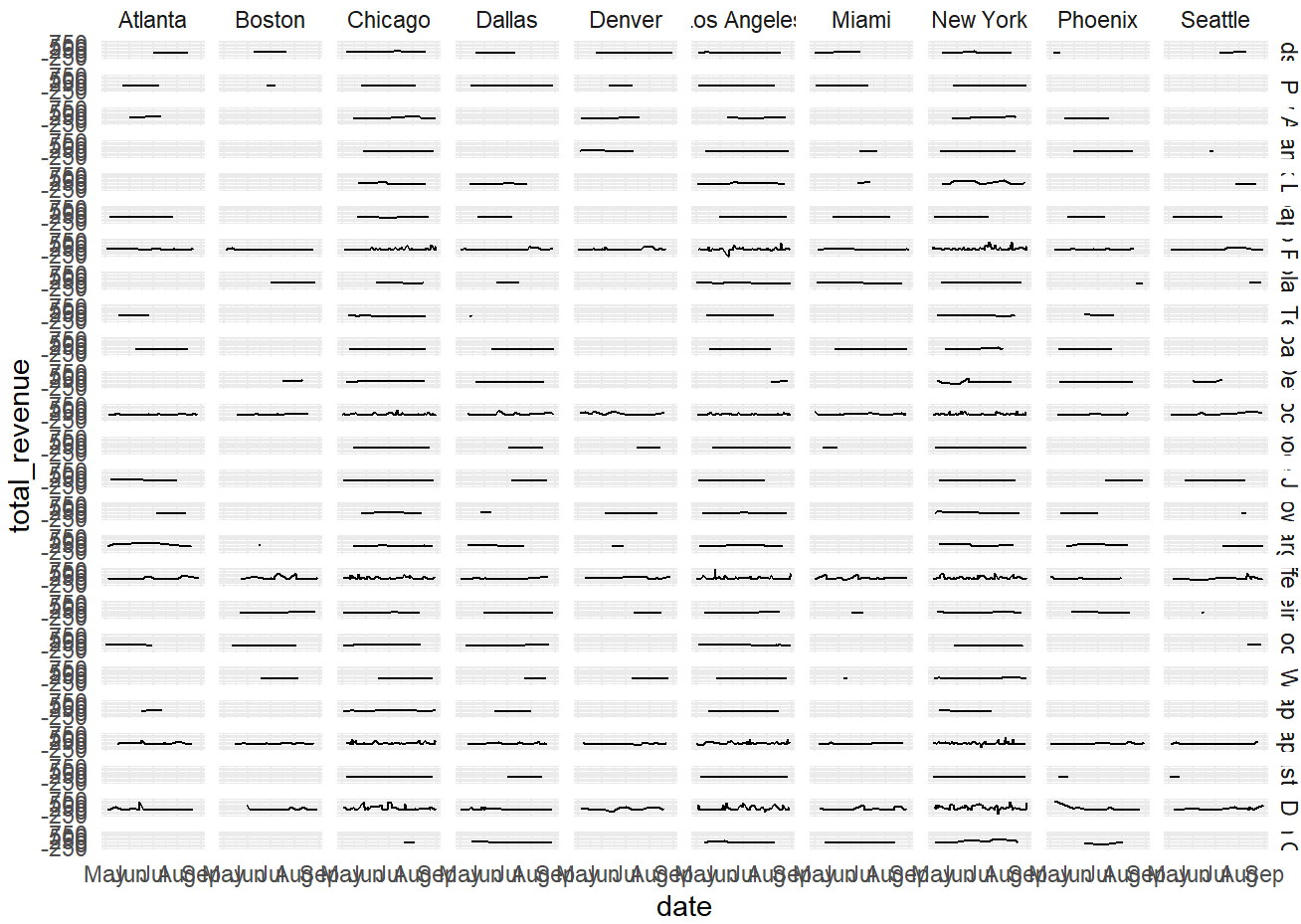
Los Angeles: Strong upward trend but high spikes in mid of the year.

Mild/Slightly Upward and Downward Trend:

Rest of the store location has higher trend except Miami and Phoenix.

Denver and Dallas was flat in overall trend with small spike in mid year.

Question: How are top products trending by Location?



Question: Which location is trending the best in terms of profit?

```
# A tibble: 10 × 5
   store_city  total_revenue total_profit total_quantity revenue_per_unit
   <fct>               <dbl>        <dbl>          <int>            <dbl>
 1 New York           43772.       20483.           3333             13.1
 2 Los Angeles        31084.       14635.           2368             13.1
 3 Chicago            30113.       14221.           2318             13.0
 4 Dallas              9580.        4603.            746             12.8
 5 Atlanta             9359.        4475.            766             12.2
 6 Boston              8428.        4043.            633             13.3
 7 Miami               7973.        3874.            622             12.8
 8 Denver              7857.        3741.            644             12.2
 9 Seattle             7471.        3645.            552             13.5
10 Phoenix             7251.        3415.            557             13.0
```

These store locations are trending by profit by generating more profit: New York
Los Angeles
Chicago
Dallas
Boston
Miami
Denver

Seattle
Phoenix

Question: Which product category is the trending the best in terms of profit?

```
# A tibble: 25 × 5
   product_name       total_revenue total_profit total_quantity revenue_per_unit
   <fct>                      <dbl>        <dbl>          <int>            <dbl>
 1 USB Flash Drive 1…        34099.       16771.           1444            23.6
 2 Premium Coffee Be…        31182.       14793.           1821            17.1
 3 Espresso Pods 20ct        28380.       10470.           2107            13.5
 4 Luxury Chocolate …        19199.       10464.           1747            11.0
 5 Toilet Paper 12pk         21415.        9697.           1674            12.8
 6 Desk Lamp                  3861.        1797.            172            22.4
 7 Phone Charger USB…         3294.        1606.            225            14.6
 8 Vitamin C 120ct            2370.        1306.            152            15.6
 9 Battery AA 12ct            1846.        1018.            207             8.92
10 Laundry Detergent…         1822.         986.            152            12.0
# ℹ 15 more rows
```

USB Flash Drive 128GB is the trending the best in terms of profit.

III. Executive Summary

A sum of $162,888 came in as total revenue, pulling a gross profit of $77,135.06 - solid returns show up clearly here. The bulk of that income ties back to just a few standout items, one alone hitting $34,098.88 at the top. Everyday picks like Ballpoint Pens 10ct bring steady gains, so do Notebook A5 units. Hand Soap 500ml turns a reliable margin, much like Toothpaste 100ml does too. Even Sparkling Water 12pk holds its ground, proving basics can deliver when priced right.

Over there where New York sits on the map, numbers climb highest - $43,771.89 pulled in, leaving others behind by a stretch. Big city energy fuels most spikes, surges, uneven swings in income across zones.

A tenth off tends to lift sales without slicing into profits, yet steeper cuts often shrink earnings even when income climbs. What holds margins together usually avoids going too far below ten percent. ## At-a-glance

- **Total revenue: $162,888**
- **Total profit (gross margin): $77,135.06**
- **Top product by revenue: $34,098.88** (0)
- **Top product by profit: Ballpoint Pens 10ct, Notebook A5, Hand Soap 500ml, Toothpaste 100ml, Sparkling Water 12pk** (0)

- **Top location by revenue:** New York ($43,771.89)

What this means

- Profit margins look solid, yet most of the earnings come from just a handful of items.
- New York, Chicago, and Los Angeles are the critical revenue sites and should be strategic focus.
- SUMMER10 and CLEARANCE are good promo codes and drives revenue upwards.
- Existing customers provide more revenue and with the revenue it also boosts the sales and profit with it.

Recommended actions

- Making sure there is enough inventory in top revenue locations like New York, Chicago and Los Angeles.
- Standardize the SUMMER10 promo code to keep the customers attracted.
- Keep the other promo code at minimum level.

IV. Communicate Findings

This is the **evidence section**. It should connect directly back to the question/problem in Step 1 and point to the specific tables/plots generated in Step 4–5.

Key findings

1. **[Revenue is highly dependent on small portion of the products]**
   - Evidence: Pareto Chart (Top 20% of products generate ~80% of revenue).
   - Interpretation: number of products are making up most of the business
2. **[Major cities make up most of the revenues]**
   - Evidence: trend by city plot \
   - Interpretation: cities has strong spikes then other cities, which indicates more revenue

Notes, assumptions, and limitations

- **[Any missing data, skew, seasonality concerns, outliers, promo effects, etc.]**
- `A handful of big deals push overall numbers up, since income patterns favor larger amounts at the high end.`
- `Most sales are low in value but the occasional high value transaction make the distribution more broad and gives more outliers which manipulate the analysis.`
- `From May through September, shifts in seasonality could affect income levels.`
- **[What you would validate next if you had more time]**
- Increase the sample size on promo codes to minimize the outliers. # V. Take Action

Turn the findings into an **action plan** that someone can execute.

Actions to take now (0–30 days)

- [Increase the stokes in top revenue cites with top revenue generating products]

— Owner: [Stock buyer]

— Metric: [Checking the quantity]

— Due: [0-30 days]

Actions to take next (30–90 days)

- [Focus on strengthening the interaction with returning customer and increasing the SUMMER10 promo code]

— Owner: [Marketing]

— Metric: [Revisiting the revenue trend by customer type and promo code]

— Due: [30-90 days]

Follow-up analysis to reduce uncertainty

- [What additional cuts / experiments / segmentations would improve confidence]
- The implication of other promo codes and less revenue generating products and location should be very selective.